

HUMBER ENGINEERING

MENG-3020

SYSTEMS MODELING & SIMULATION

LECTURE 8

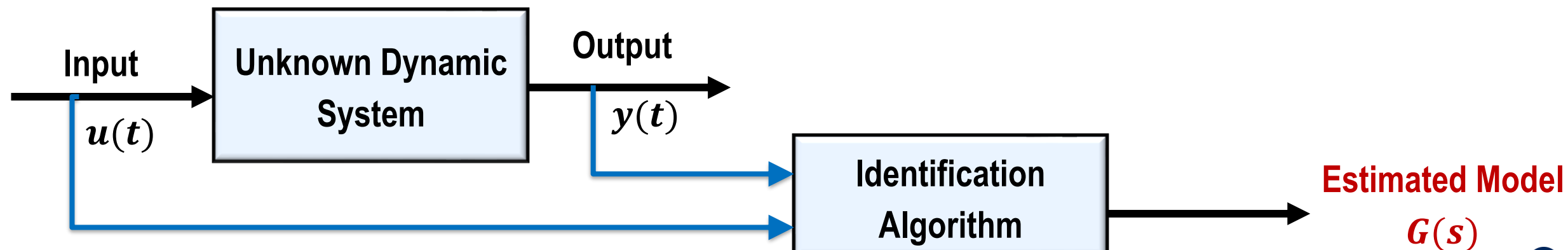
LECTURE 8

System Analysis in Time Domain

- Introduction
- Time Response of First-Order Systems
 - Transient Response
 - System Modeling from Step Response
- Time Response of Second-Order Systems
 - Transient Response
 - System Modeling from Step Response
- Stability of Systems
- Low-Order Models Approximation

Introduction

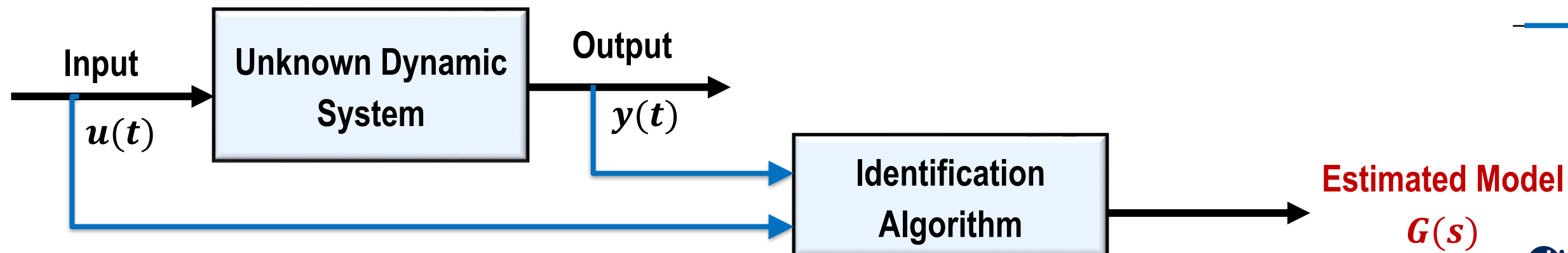
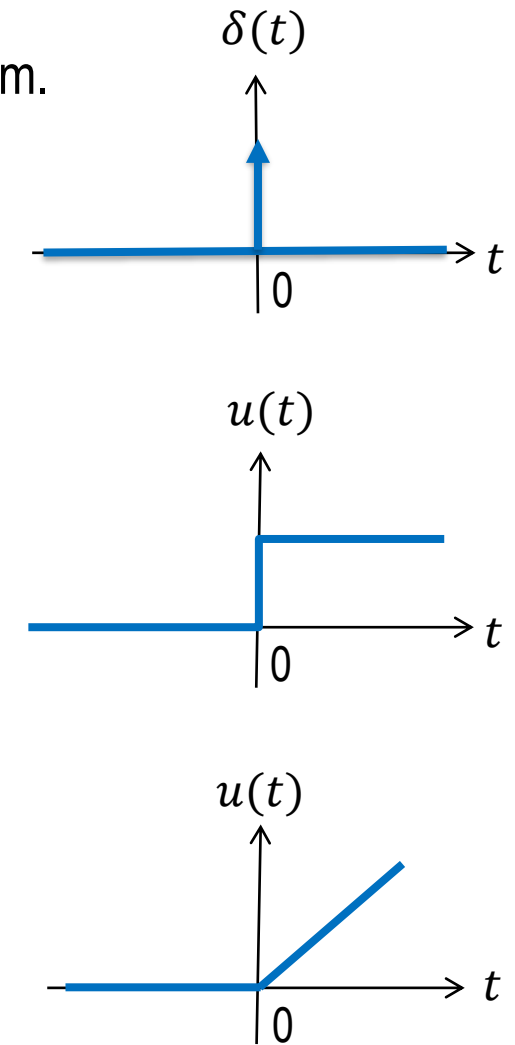
- **System identification** refers to obtaining the **transfer function** $G(s)$ of a system by only considering its **output response** to a given particular **input signal**.
- This is useful when we have not much information about a system at hand.
- For example, it is a **black box** or **too complicated** to be modeled, and we need to find out its transfer function for simulation or control system design purposes.
- Although we assume that no information is provided by the system, some hypothesis must be considered, like for instance the **order** of the system.
- In this course we will focus on system identification and transfer function modeling via
 - **Transient response analysis** of **first-order** and **second-order** systems. (**Traditional graphical techniques**)
 - **Modern system identification methods** for **black-box** system identification. (**Statistical methods**)



Introduction

□ Transient Response Analysis

- The **goal** is understanding system behavior in **time-domain**, in particular, the **transient response** of the system.
- The commonly use input signals to test a system's response in the **time-domain** are:
 - **Impulse Function**: To model suddenly applied and suddenly removed input
 - **Step Function**: To model a suddenly applied input that remains constant
 - **Ramp Function**: To model an input that is changing at a constant rate
- Among those inputs, **step response** is more appropriate for transient response analysis, it can provide lots of information regarding the characteristics and transfer function model of the system.
- Since most of the physical systems can be modeled or *approximated* as a **first-order** or **second-order** transfer functions, our focus will be on characterizing those systems.



First-Order Systems

Transfer Function Model

- First-order systems are described by a first-order differential equation.

Differential Equation $\rightarrow \tau y'(t) + y(t) = Ku(t)$

- The standard form of a first-order transfer function model is:

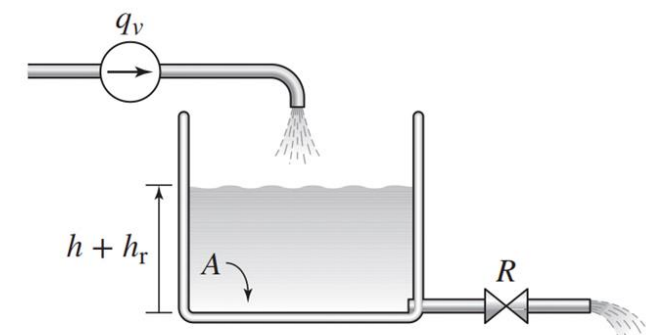
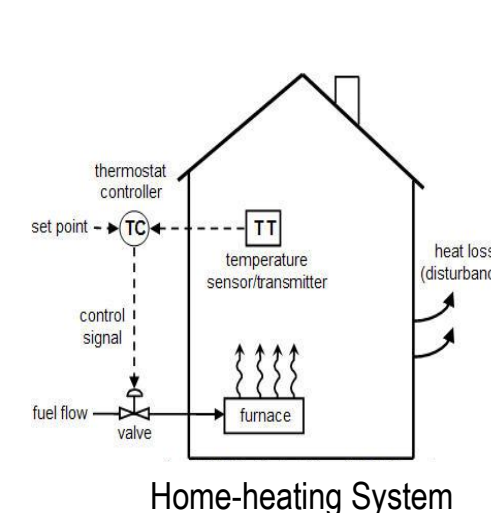
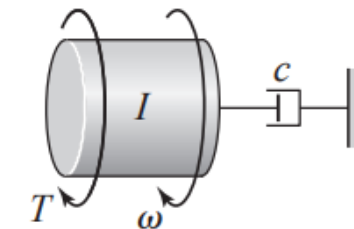
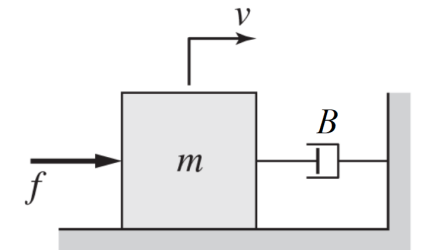
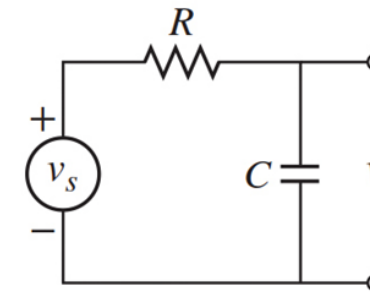
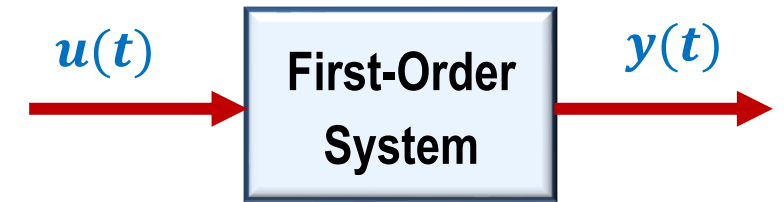
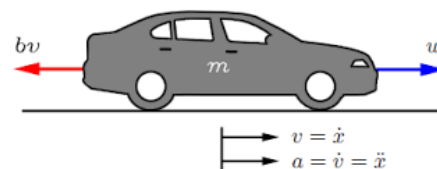
$$G(s) = \frac{Y(s)}{U(s)} = \frac{K}{\tau s + 1}$$

Steady-state gain
Static gain
DC-gain

Time constant

- First-order systems have a single energy-storage element.

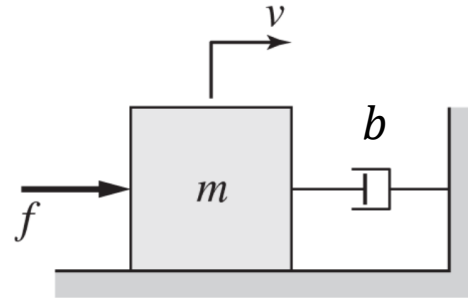
- RC or RL Electric Circuits
- Single-Tank Liquid Level System
- Mass-damper Systems
- Single-capacity Thermal Systems



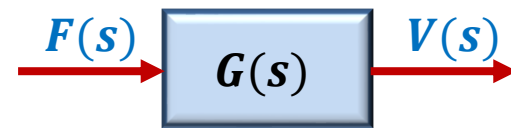
First-Order Systems – Example

$$G(s) = \frac{K}{\tau s + 1}$$

- Simple examples of systems that can be modeled as a **first-order** system.



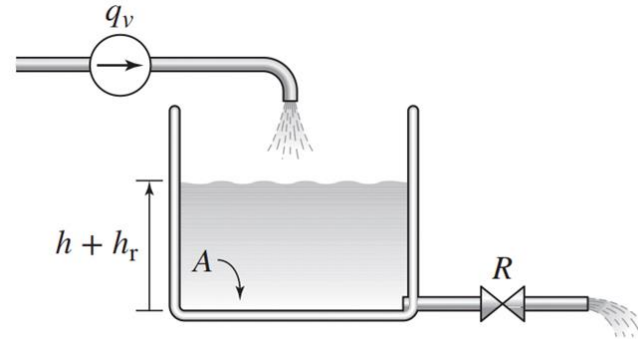
$$m\dot{v}(t) + bv(t) = f(t)$$



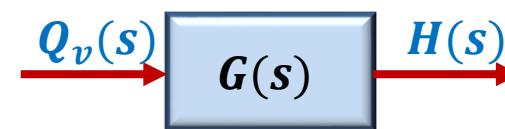
$$G(s) = \frac{V(s)}{F(s)} = \frac{1}{ms + b} = \frac{\frac{1}{b}}{\frac{m}{b}s + 1}$$

- Time constant & DC-gain

$$\tau = \frac{m}{b}, \quad K = \frac{1}{b}$$



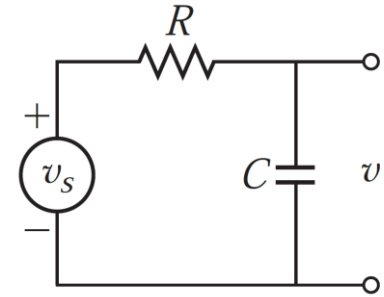
$$AR\dot{h}(t) + gh(t) = Rq_v(t)$$



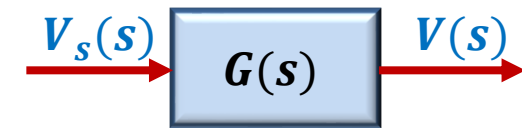
$$G(s) = \frac{H(s)}{Q_v(s)} = \frac{R}{ARs + g} = \frac{\frac{R}{g}}{\frac{AR}{g}s + 1}$$

- Time constant & DC-gain

$$\tau = \frac{AR}{g}, \quad K = \frac{R}{g}$$



$$RC\dot{v}(t) + v(t) = v_s(t)$$



$$G(s) = \frac{V(s)}{V_s(s)} = \frac{1}{RCs + 1}$$

- Time constant & DC-gain

$$\tau = RC, \quad K = 1$$

First-Order Systems

□ Step Response

- **Unit-step response** of a **first-order** system is determined as,

$$Y(s) = G(s)U(s) = \left(\frac{K}{\tau s + 1} \right) \left(\frac{1}{s} \right) = \frac{K}{s(\tau s + 1)} = \frac{K}{s} + \frac{-K\tau}{\tau s + 1}$$

- Taking inverse **Laplace transform** gives the **unit-step response**,

$$y(t) = K - Ke^{-t/\tau}, \quad t \geq 0$$

Unit-Step Response

Steady-State Response

y_{ss}

Transient Response

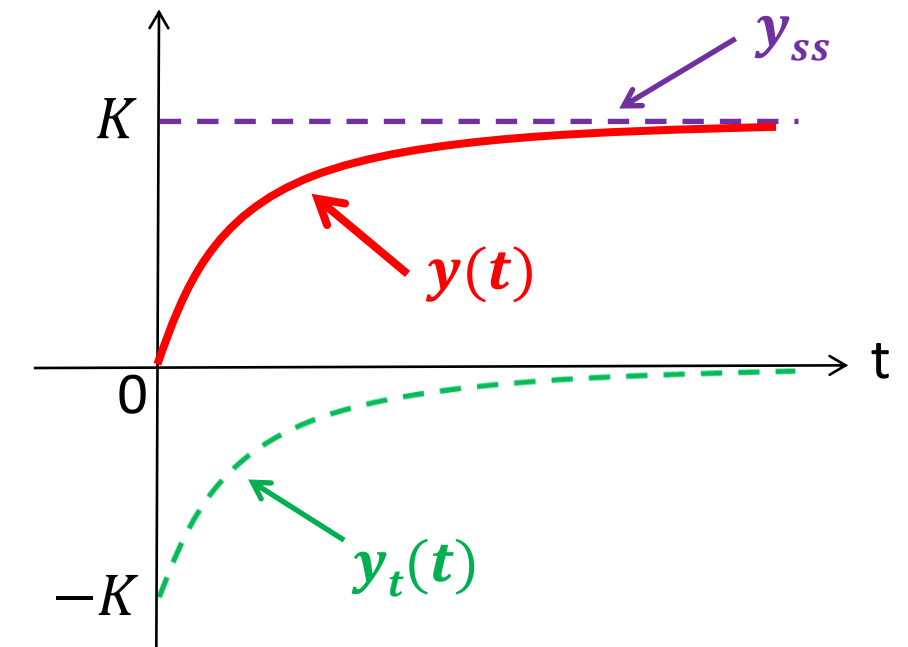
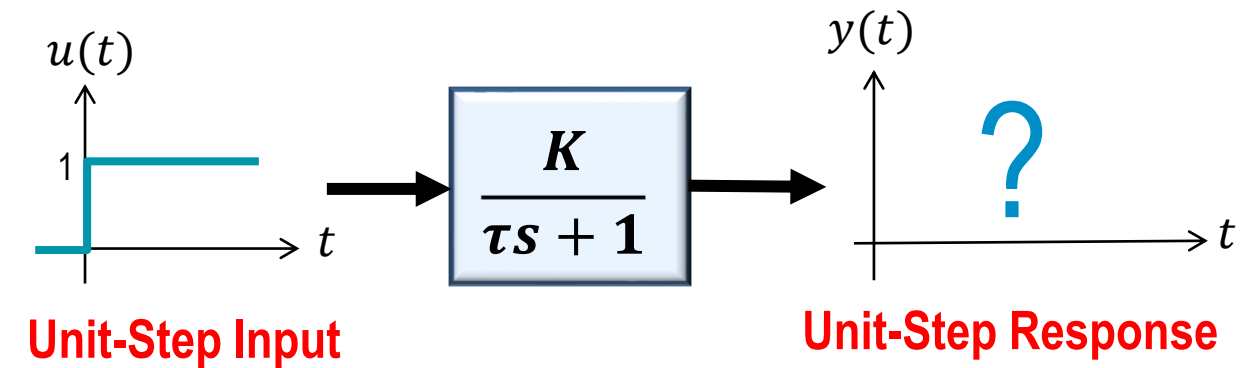
$y_t(t)$

- **DC-gain** shows final value of the unit-step response in a **stable** system.

$$y_{ss} = y(\infty) = \lim_{t \rightarrow \infty} y(t) = K$$

- **DC-gain** is also determined from **Final-Value Theorem**:

$$y_{ss} = y(\infty) = \lim_{s \rightarrow 0} sY(s) = \lim_{s \rightarrow 0} sG(s)U(s) = \lim_{s \rightarrow 0} sG(s) \left(\frac{1}{s} \right) = \lim_{s \rightarrow 0} G(s)$$



First-Order Systems

□ Properties of Step Response

- The **step-response** of a **first-order** system is an **exponential curve**.

$$y(t) = K - Ke^{-t/\tau}, \quad t \geq 0$$

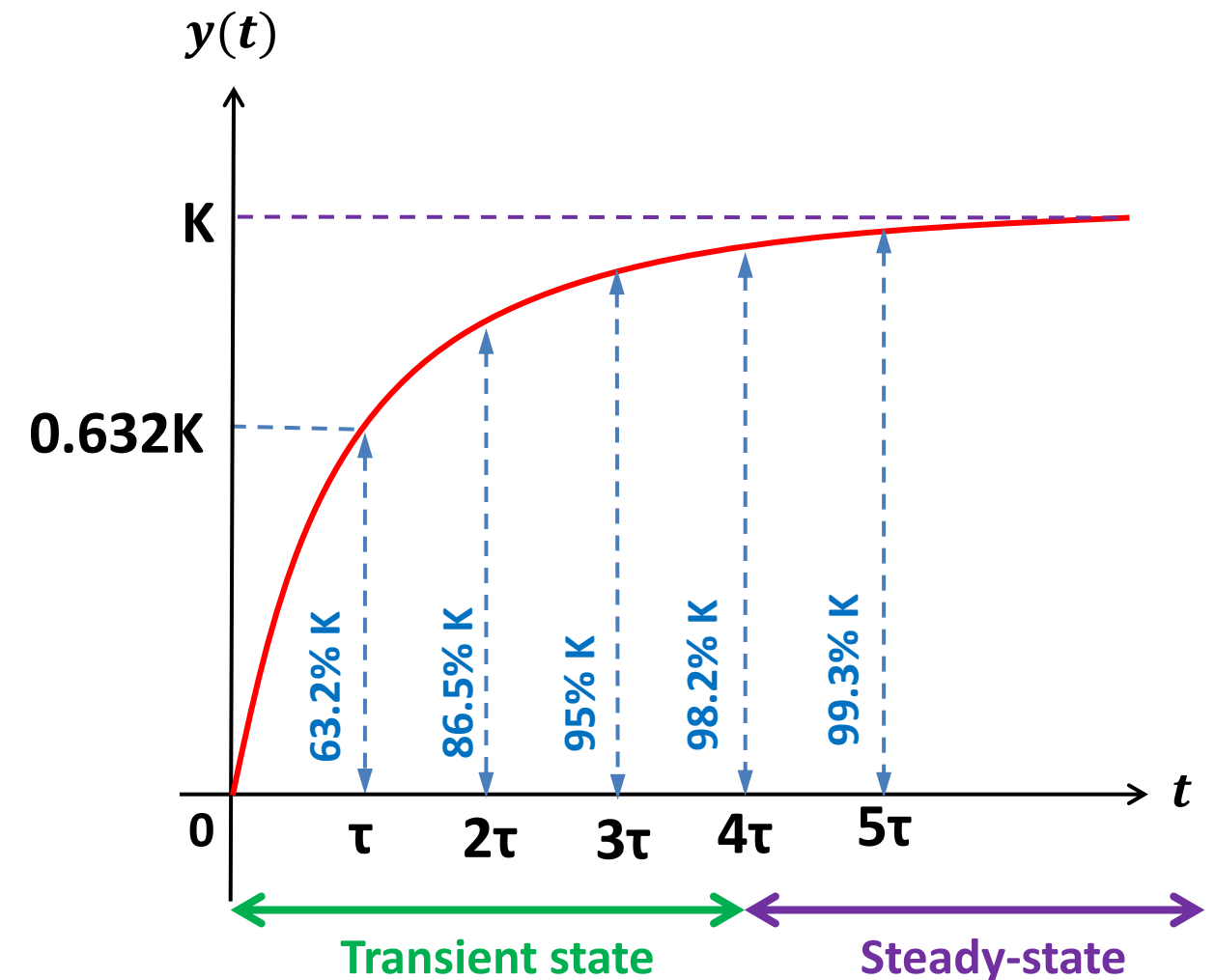
- Time constant** is defined as the time when the unit-step response $y(t)$ has reached **63.2%** of its total change from its **initial** value to the **steady-state** value.

At $t = \tau \rightarrow y(\tau) = K(1 - e^{-\tau/\tau}) = K(1 - e^{-1}) = 0.632K$

- Time-Constant** shows how fast a first-order system responds to the step input. **Smaller** time constant means **faster** response.
- The **transient state** and the **steady-state state** are distinguished by the **Settling Time (t_s)**.
- It is the required time for the step response to reach and stay within the specified percentage of its final value (**usually 2% criteria**)

$$\bullet \text{ 2\% criteria } \rightarrow t_s = 4\tau$$

$$G(s) = \frac{K}{\tau s + 1}$$



First-Order Systems – Example

Example 1

Consider the following electrical network with input $v(t)$ and output $v_c(t)$

a) Determine the differential equation and transfer function representing the dynamic model of the system.

Applying the Kirchhoff's Voltage Law (KVL) we have

$$v(t) = v_R(t) + v_c(t) \rightarrow v(t) = Ri(t) + v_c(t)$$

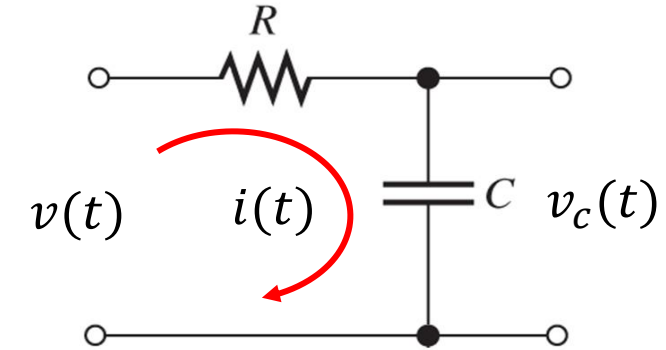
The differential equation relating $v(t)$ to $v_c(t)$ is determined as

$$\xrightarrow{i(t) = C \frac{dv_c(t)}{dt}}$$

$$v(t) = RC \frac{dv_c(t)}{dt} + v_c(t)$$

First-order differential equation

$$G(s) = \frac{V_c(s)}{V(s)}$$



Taking the Laplace transform (zero initial conditions, $v_c(0) = 0$)

$$V(s) = RCsV_c(s) + V_c(s)$$

$$G(s) = \frac{V_c(s)}{V(s)} = \frac{1}{RCs + 1}$$

First order transfer function

b) Determine the transfer function, the steady-state gain and the time constant for $R = 100\Omega$, $C = 0.05F$.

$$G(s) = \frac{1}{5s + 1} \rightarrow \begin{cases} \text{Steady-state gain} \rightarrow K = 1 \\ \text{Time constant} \rightarrow \tau = RC = 5 \text{ sec} \end{cases}$$

$$G(s) = \frac{K}{\tau s + 1}$$

First-Order Systems – Example

Example 1

Consider the following electrical network with input $v(t)$ and output $v_c(t)$

c) If the applied voltage is $v(t) = 6\text{ V}$, determine and plot the response of the system if the initial voltage of the capacitor is zero, $v_c(0) = 0$.

Find the output response by applying the partial fraction decomposition and taking the inverse Laplace transform:

$$G(s) = \frac{V_c(s)}{V(s)} = \frac{1}{5s + 1} \longrightarrow V_c(s) = G(s)V(s) = \left(\frac{1}{5s + 1} \right) \left(\frac{6}{s} \right)$$

$$V_c(s) = \frac{6}{s} + \frac{-30}{5s + 1} = \frac{6}{s} - \frac{6}{s + 1/5}$$

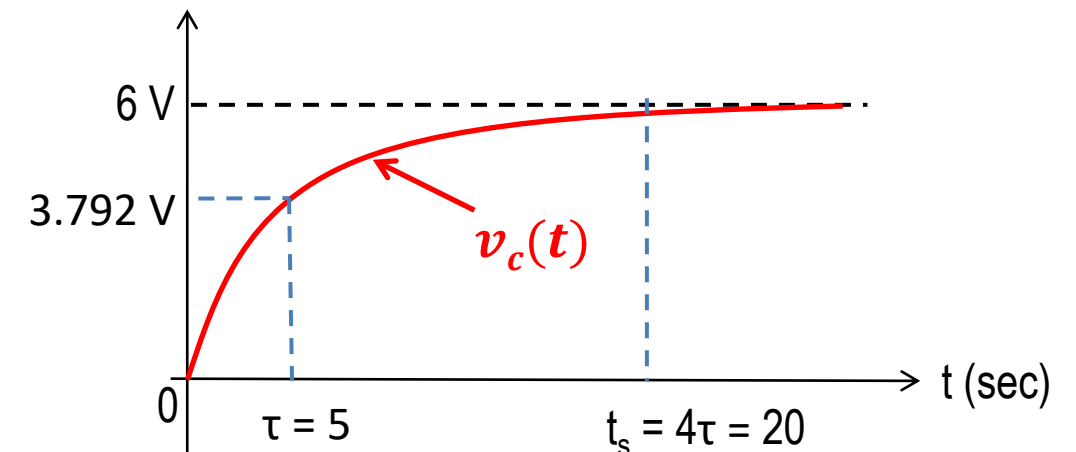
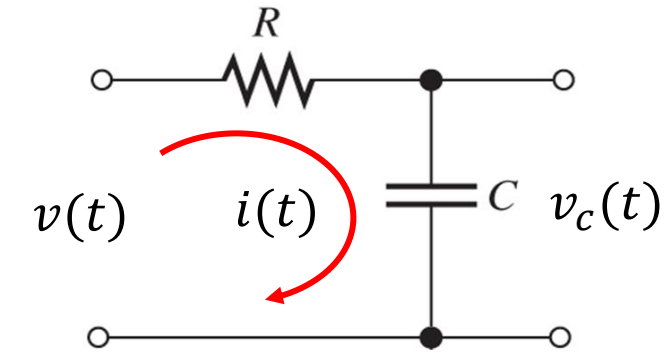
$$v_c(t) = 6 - 6e^{-t/5}, \quad t \geq 0$$

Step Response

Initial value $\rightarrow \lim_{t \rightarrow 0} v_c(t) = 0$

Final value $\rightarrow \lim_{t \rightarrow \infty} v_c(t) = 6$

Settling time $\rightarrow t_s = 4\tau = 4RC = 20\text{ sec}$



First-Order Systems – Example

Example 1

Consider the following electrical network with input $v(t)$ and output $v_c(t)$

d) If the applied voltage is $v(t) = 1\text{ V}$, determine and plot the response of the system if the initial voltage of the capacitor is zero, $v_c(0) = 0.1\text{ V}$.

Having the differential equation model from Part (a), take Laplace transform by considering the initial conditions:

$$v(t) = RC\dot{v}_c(t) + v_c(t) \rightarrow V(s) = RC(sV_c(s) - v_c(0)) + V_c(s)$$

$$V(s) = (RCs + 1)V_c(s) - RCv_c(0) \rightarrow V_c(s) = \frac{1}{RCs + 1}V(s) + \frac{RC}{RCs + 1}v_c(0)$$

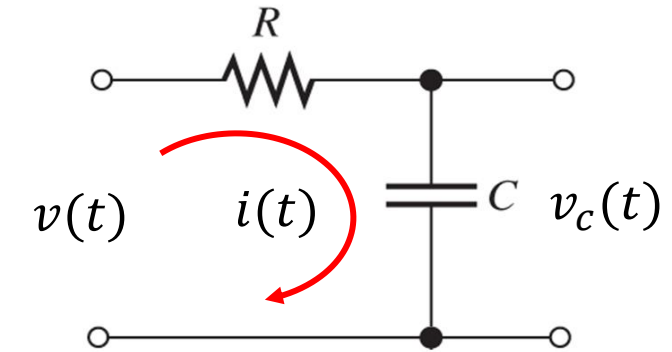
$$V_c(s) = \left[\frac{1}{s(5s + 1)} \right] + \left[\frac{0.5}{5s + 1} \right] = \left[\frac{1}{s} + \frac{-5}{5s + 1} \right] + \left[\frac{0.5}{5s + 1} \right] \rightarrow v_c(t) = \underbrace{(1 - e^{-t/5})}_{\text{Forced Response}} + \underbrace{(0.1e^{-t/5})}_{\text{Free Response}}$$

$$v_c(t) = 1 - 0.9e^{-t/5}, \quad t \geq 0 \quad \text{Step Response}$$

Initial value $\rightarrow \lim_{t \rightarrow 0} v_c(t) = 0.1$

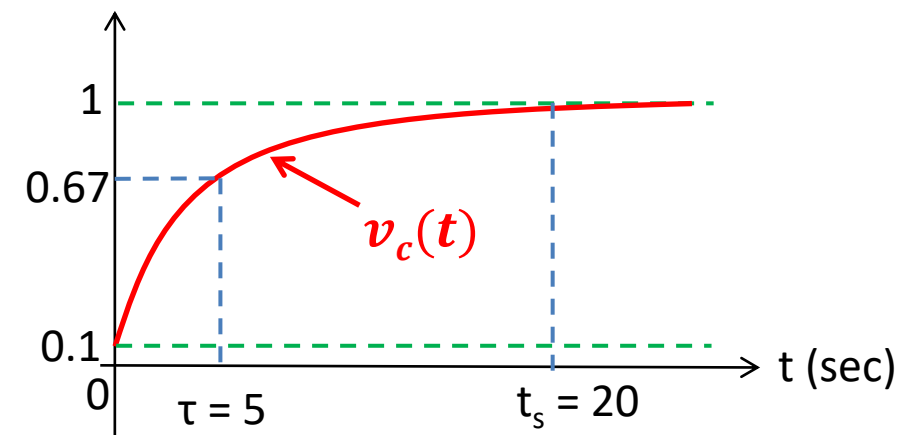
Final value $\rightarrow \lim_{t \rightarrow \infty} v_c(t) = 1$

Settling time $\rightarrow t_s = 4\tau = 4RC = 20\text{ sec}$



Forced Response
Zero-initial-conditions Response

Free Response
Zero-input Response

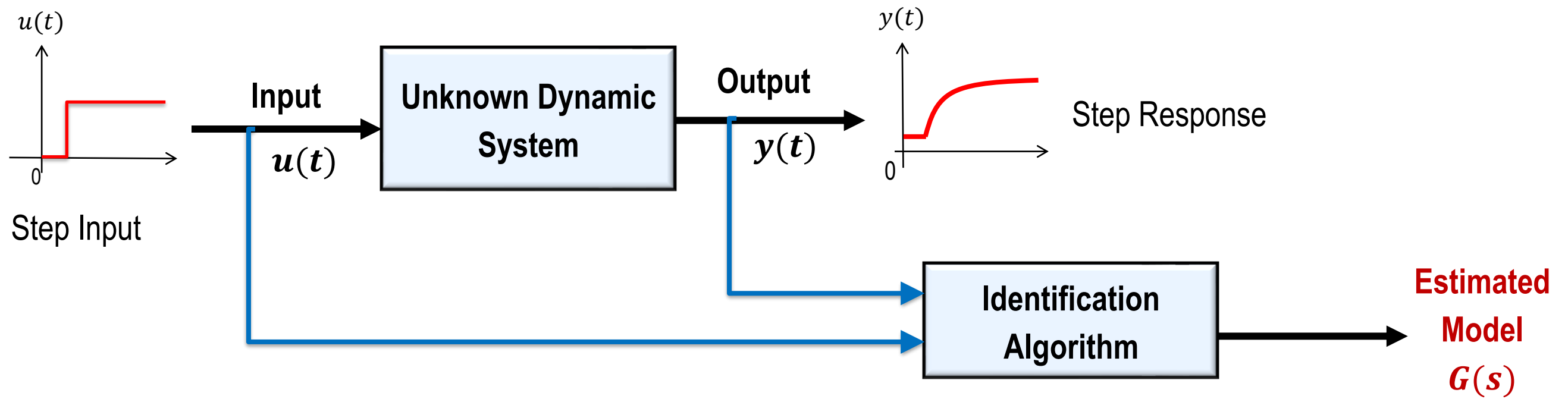


First-Order Model Identification from Step Response

- First-order models are characterized by their time-constant τ and steady-state gain K .

$$G(s) = \frac{K}{\tau s + 1}$$

- The model parameters can be identified by transient response analysis of the step response of the system.
- The methods are based on the experimental graphical techniques.



First-Order Model Identification from Step Response

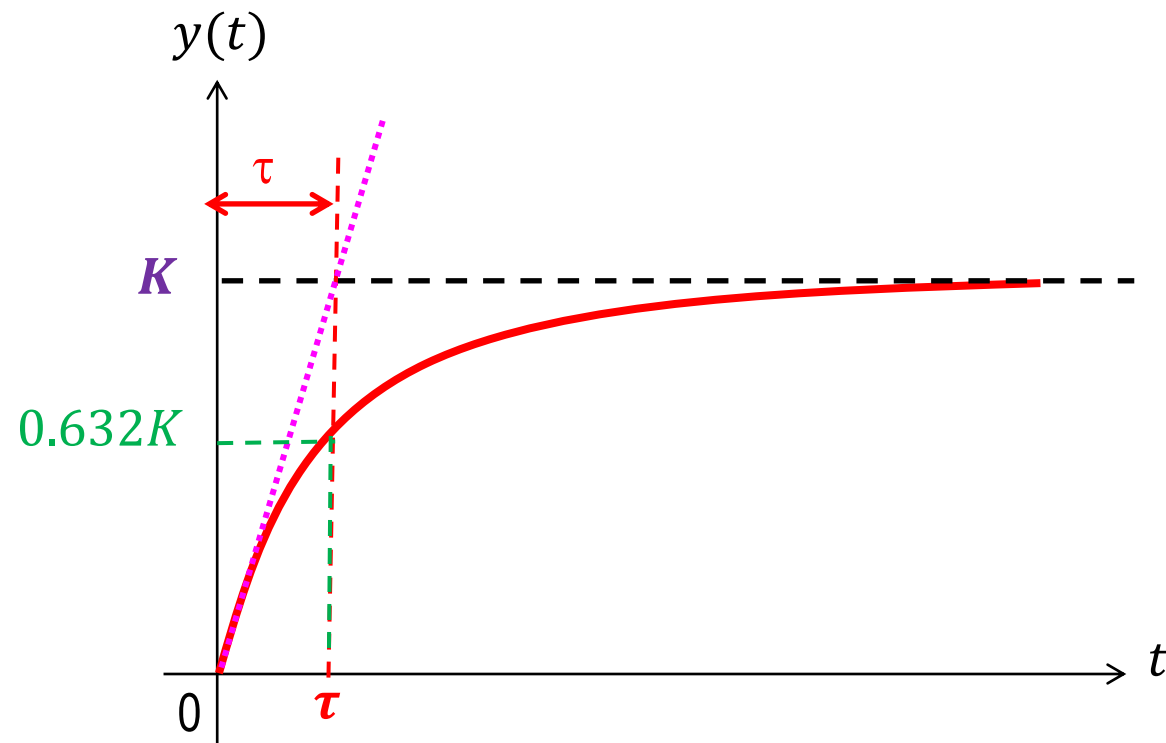
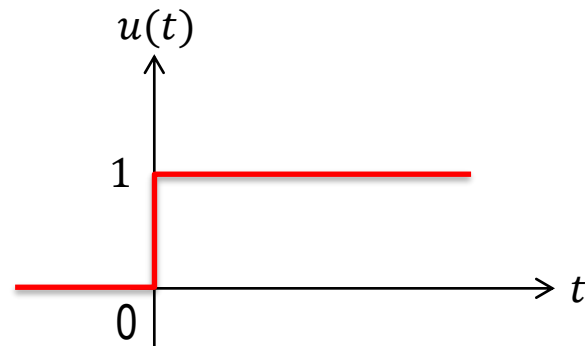
□ The Method of 63% Change

- The **time-constant** τ and **steady-state gain** K are determined based on their definitions from the step response.

$$G(s) = \frac{K}{\tau s + 1}$$

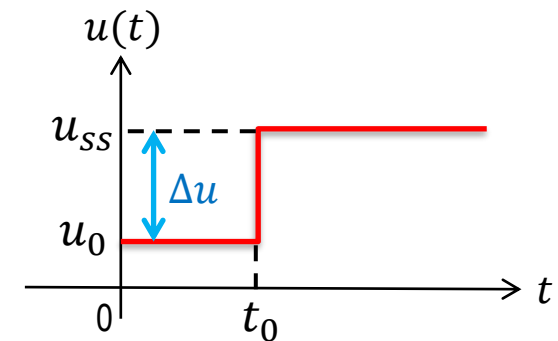
• Unit-step Response

$$u(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases}$$

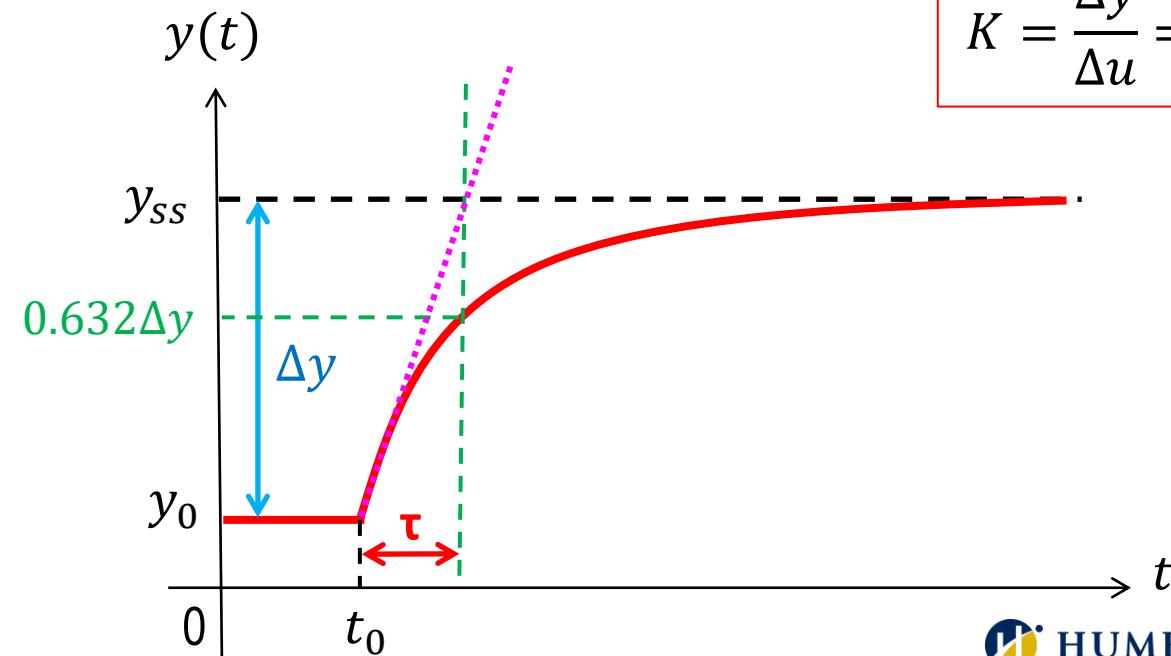


• General Step Response

$$u(t) = \begin{cases} u_{ss} & t \geq t_0 \\ u_0 & 0 \leq t < t_0 \\ 0 & t < 0 \end{cases}$$



$$K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0}$$



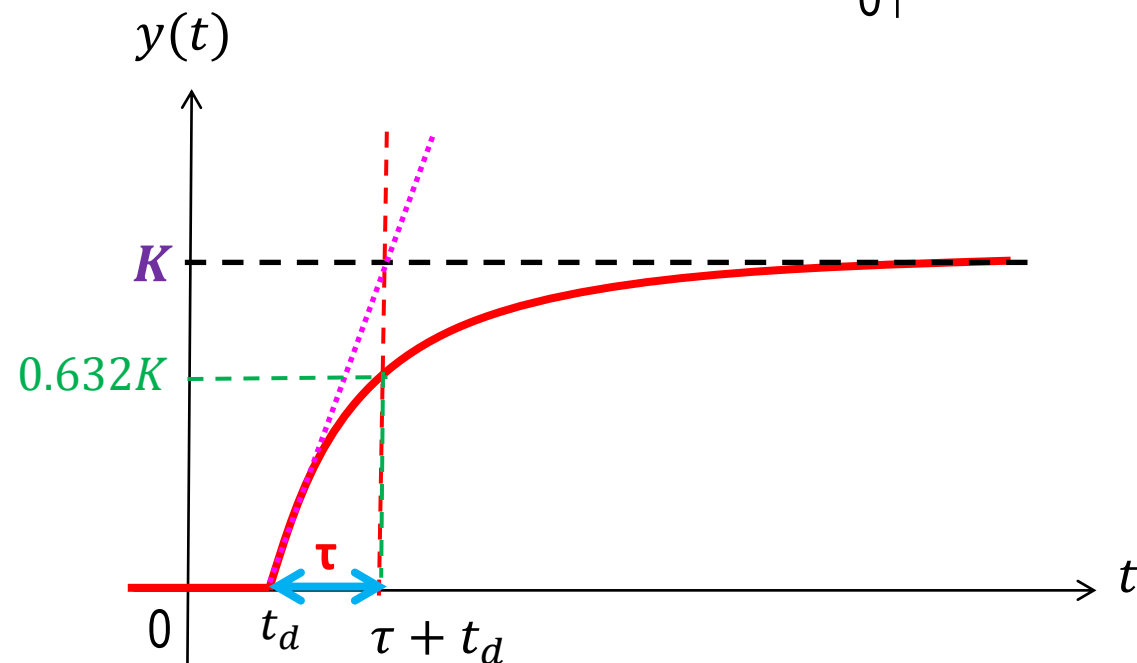
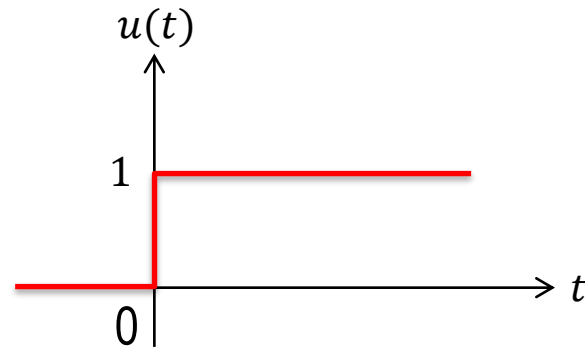
First-Order Model Identification from Step Response

□ First-Order Model with Delay Time

- In practice, a system often contains a **time-delay**.
 - A **first-order** system with a **time-delay** t_d has the transfer function of:

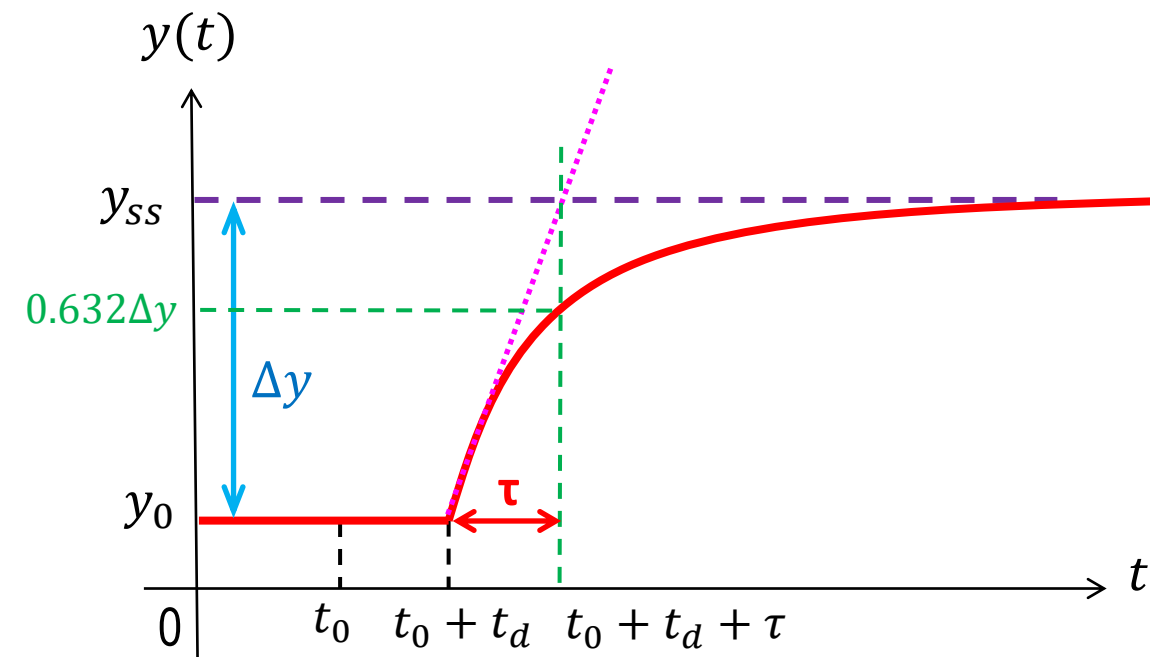
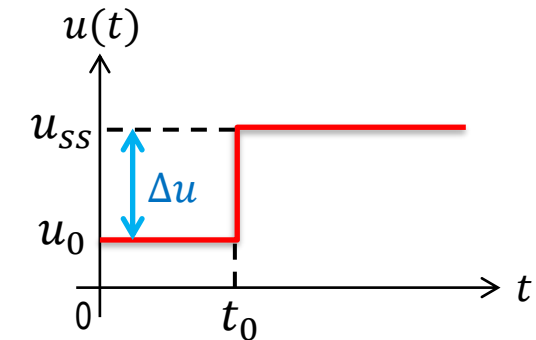
$$G(s) = \frac{K e^{-t_d s}}{\tau s + 1}$$

• Unit-step Response



• General Step Response

$$K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0}$$



First-Order Model Identification from Step Response

□ The Sundaresan-Krishnaswamy (S-K) Method

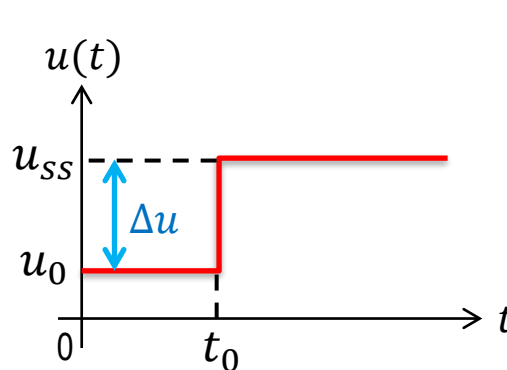
$$G(s) = \frac{K e^{-t_d s}}{\tau s + 1}$$

- In practice, a system is hardly ever a perfect first-order system.
 - Often the step response does **not** have its **steepest slope immediately at the beginning** of the response.
 - This means that the system is of **higher order than first-order**.
- In this case, we can **approximate** the system as a **first-order system with a time delay**.

• General Step Response

- The **static gain** is calculated as before:

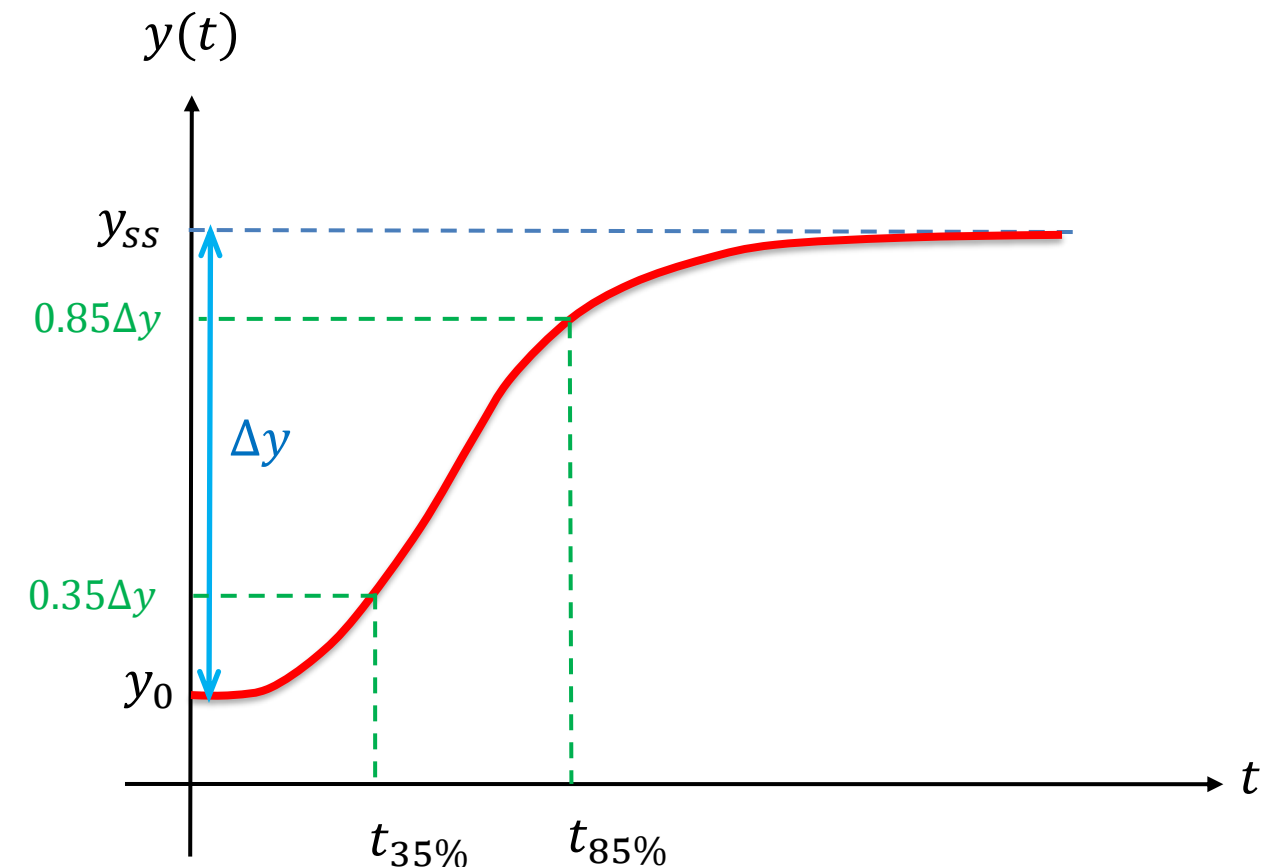
$$K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0}$$



- The **time constant** and **time delay** are estimated by:

$$\tau = 0.682(t_{85\%} - t_{35\%})$$

$$t_d = t_{35\%} - 0.431\tau$$



First-Order Model Identification – Example

Example 2

Estimate a FOPDT model for a third-order system based on the given step response, if the step input is $u(t) = 2$.

$$G(s) = \frac{300}{(s + 3)(s + 5)(s + 7)}$$

$$G(s) = \frac{Ke^{-t_d s}}{\tau s + 1}$$

We can *approximate* the system as a FOPDT model using the S-K method.

Calculate the static gain:

$$K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0} = \frac{5.71 - 0}{2 - 0} = 2.855$$

Estimate the time constant and time delay:

$$y_{85\%} = 5.71 \times 0.85 = 4.8535 \rightarrow t_{85\%} = 1.08 \text{ sec}$$

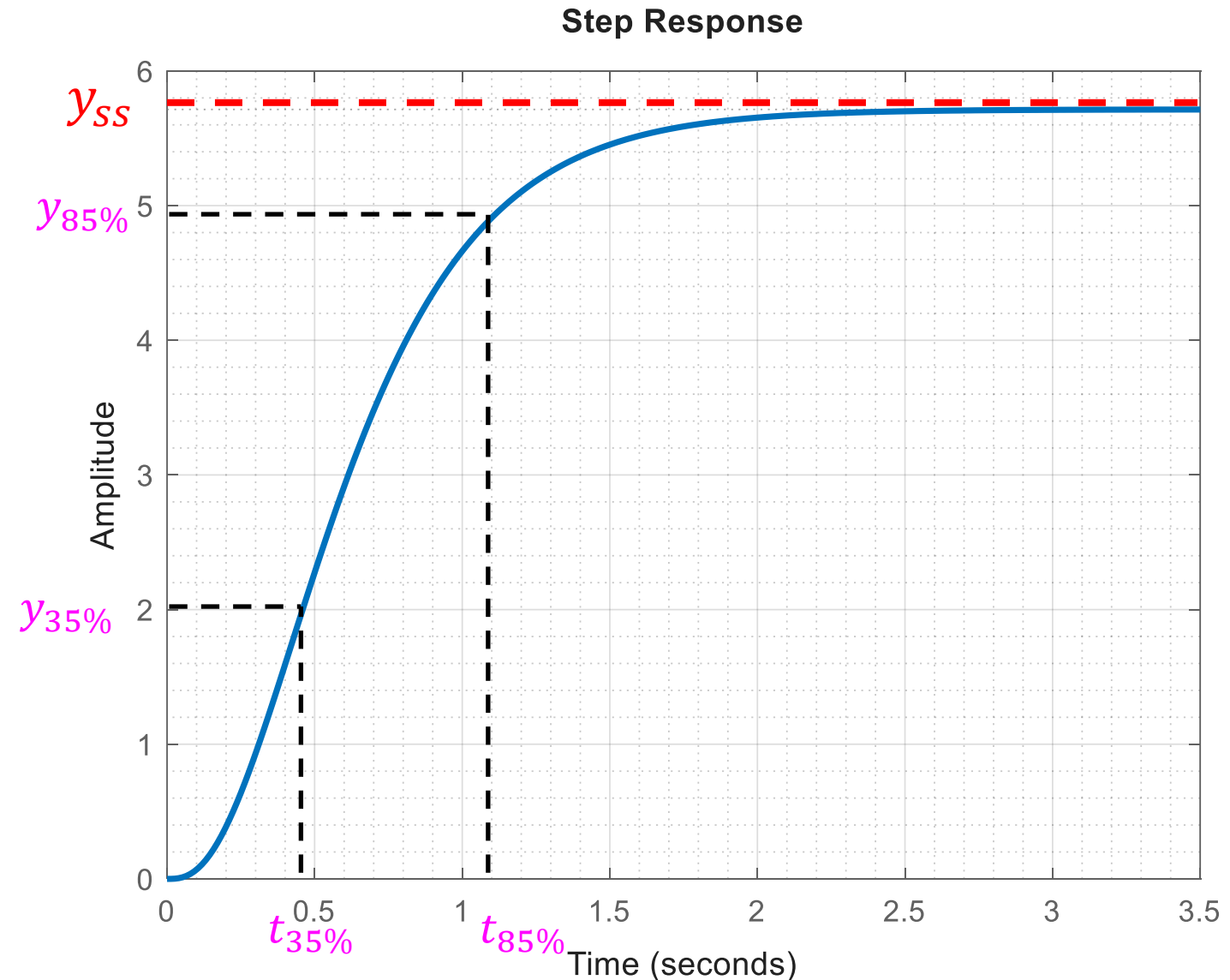
$$y_{35\%} = 5.71 \times 0.35 = 1.9985 \rightarrow t_{35\%} = 0.45 \text{ sec}$$

$$\tau = 0.682(t_{85\%} - t_{35\%}) = 0.682(1.08 - 0.45) = 0.43 \text{ sec}$$

$$t_d = t_{35\%} - 0.431\tau = 0.45 - 0.431(0.43) = 0.26 \text{ sec}$$

The estimated FOPDT model is:

$$G(s) = \frac{2.855e^{-0.26s}}{0.43s + 1}$$



First-Order Model Identification – Example

Example 2

Estimate a FOPDT model for a third-order system based on the given step response, if the step input is $u(t) = 2$.

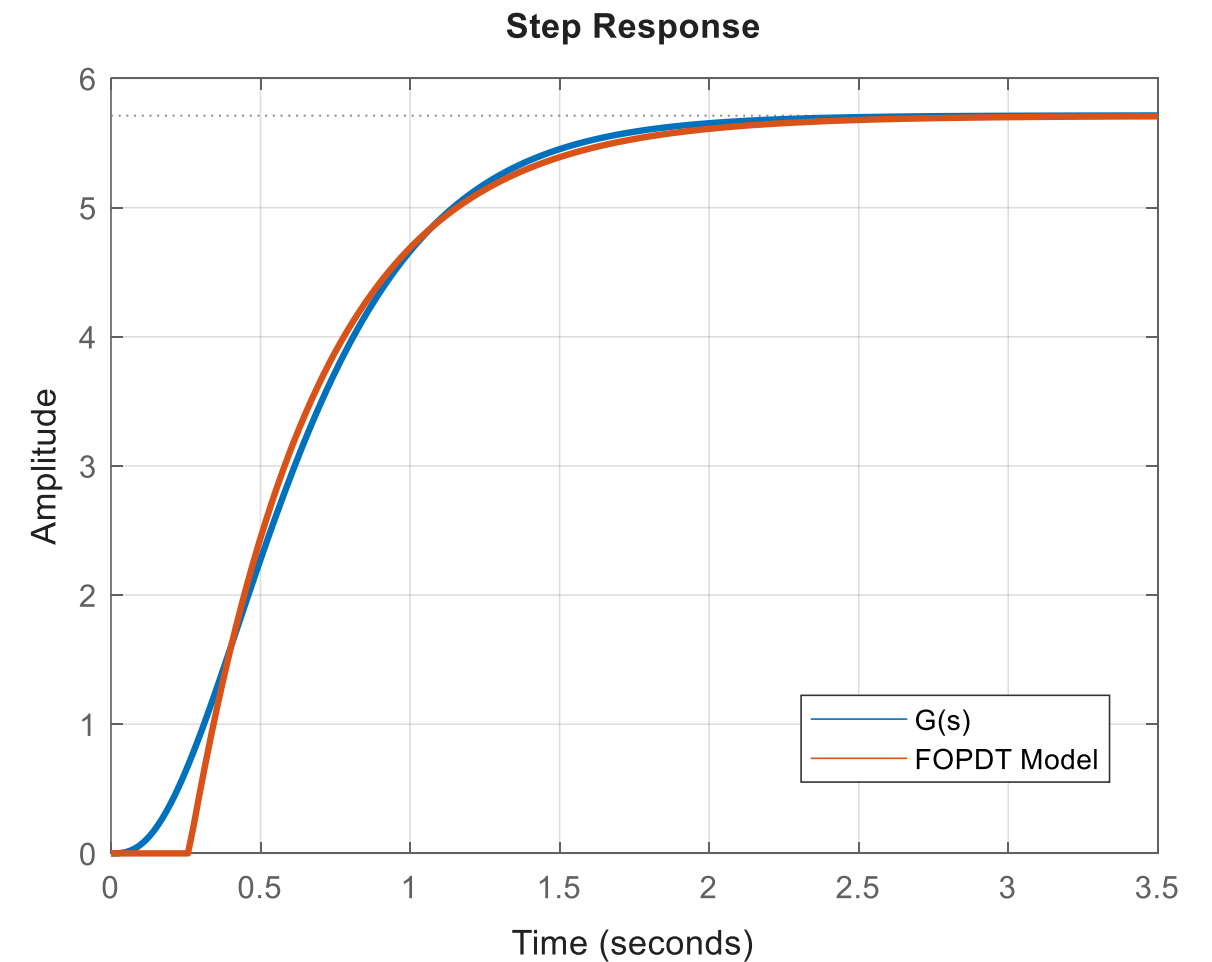
$$G(s) = \frac{Ke^{-t_d s}}{\tau s + 1}$$

$$G(s) = \frac{2.855e^{-0.26s}}{0.43s + 1}$$

$$G(s) = \frac{300}{(s + 3)(s + 5)(s + 7)}$$

Model Verification: We can plot unit-step responses of $G(s)$ and its approximated FOPDT model by MATLAB to compare them.

```
num1 = [300];  
den1 = poly([-3,-5,-7]);  
sys1 = tf(num1,den1);  
  
num2 = [2.855];  
den2 = [0.43 1];  
sys2 = tf(num2,den2,'OutputDelay', 0.26);  
  
figure;  
opt = RespConfig;  
opt.Amplitude = 2;  
step(sys1,opt), hold on  
step(sys2,opt)
```



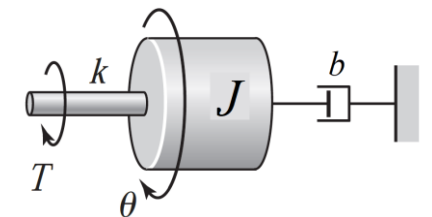
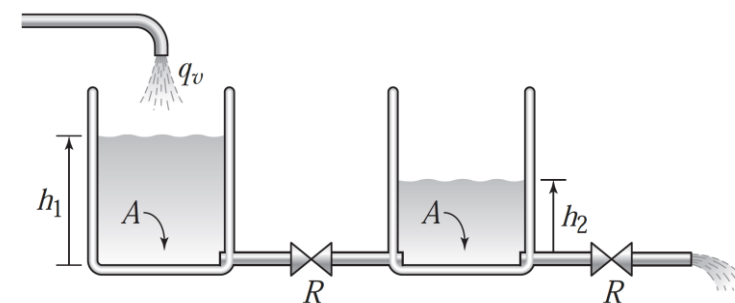
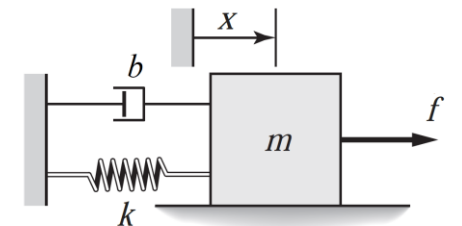
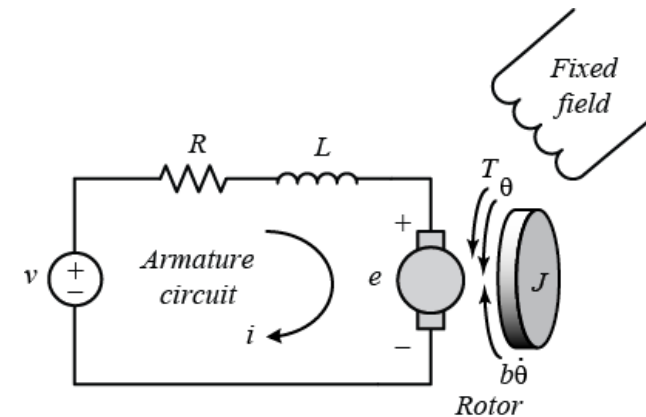
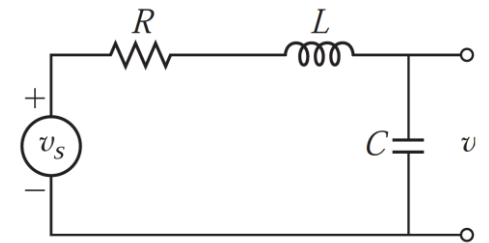
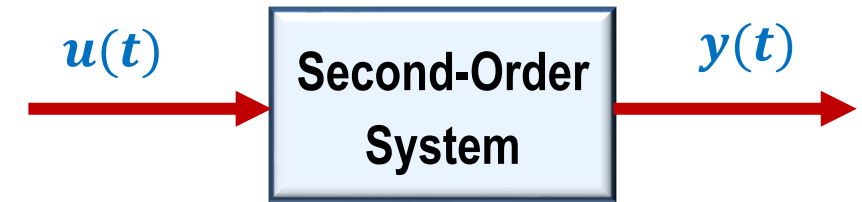
Second-Order Systems

Transfer Function Model

- **Second-order** systems are systems whose input-output relationship is a second-order differential equation.
- The standard form of a second-order transfer function model is:

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- ω_n : Undamped Natural Frequency
- ζ : Damping Ratio
- K : steady-state gain
- **Second-order** systems have two energy-storage elements.
 - RLC Electric Circuits
 - Two-Tank Liquid Level System
 - DC Motor Speed Model
 - Mass-Spring-Damper System

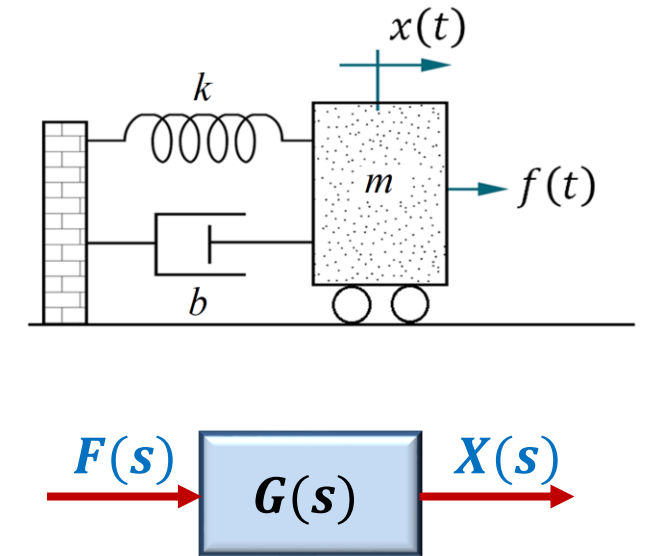


Second-Order Systems – Example

- Simple examples of systems can be modeled as a **second-order** system.

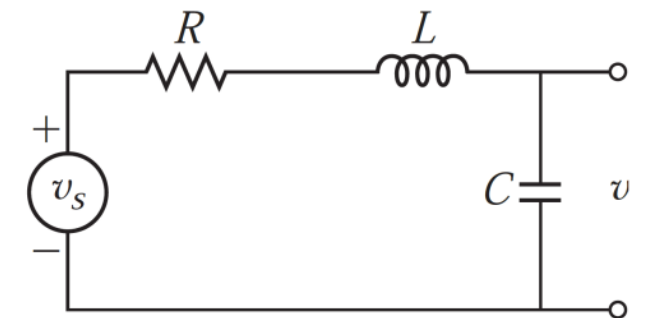
$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t) \quad \longrightarrow \quad G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + bs + k}$$

$$G(s) = \frac{\frac{1}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}} \quad \longrightarrow \quad \omega_n = \sqrt{\frac{k}{m}}, \quad \zeta = \frac{b}{2\sqrt{mk}}, \quad K = \frac{1}{k}$$



$$LC\ddot{v}(t) + RC\dot{v}(t) + v(t) = v_s(t) \quad \longrightarrow \quad G(s) = \frac{V(s)}{V_s(s)} = \frac{1}{LCs^2 + RCs + 1}$$

$$G(s) = \frac{\frac{1}{LC}}{s^2 + \frac{R}{L}s + \frac{1}{LC}} \quad \longrightarrow \quad \omega_n = \frac{1}{\sqrt{LC}}, \quad \zeta = \frac{R}{2}\sqrt{\frac{C}{L}}, \quad K = 1$$



$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Second Order Systems

□ Step Response

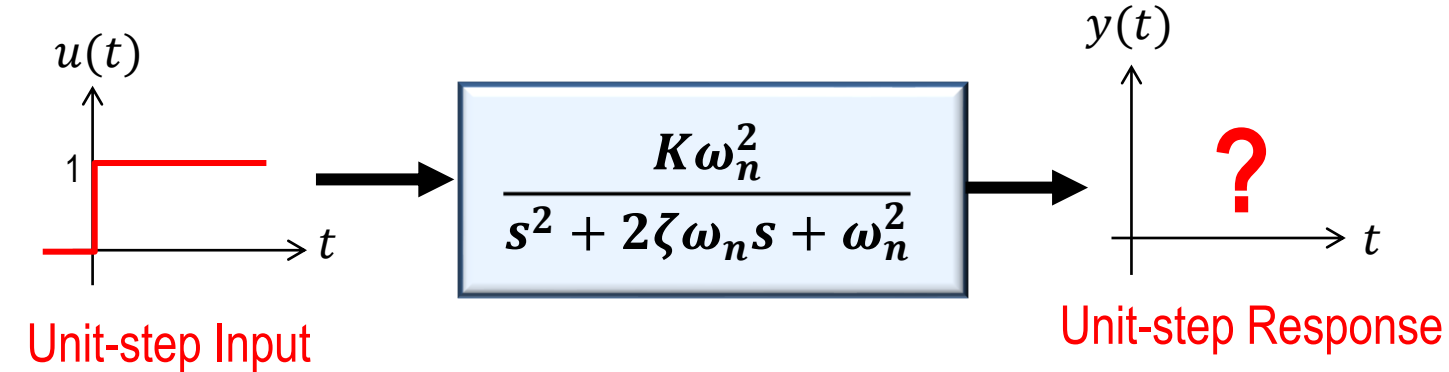
- Unit-step response of a second-order system is determined as below.

$$Y(s) = G(s)U(s)$$

$$Y(s) = \left(\frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \right) \left(\frac{1}{s} \right) = \frac{K\omega_n^2}{s(s - s_1)(s - s_2)} = \frac{A}{s} + \frac{B}{s - s_1} + \frac{C}{s - s_2} \quad \rightarrow \quad y(t) = \mathcal{L}^{-1}[Y(s)]$$

where A , B and C are determined from partial fraction expansions method.
The values depend on the pole locations s_1 and s_2 .

- The pole locations and the step response of the second-order system can be characterized based on the damping ratio ζ
 - Overdamped response if $\zeta > 1$
 - Critically-damped response if $\zeta = 1$
 - Underdamped response if $0 < \zeta < 1$
 - Undamped response if $\zeta = 0$
 - Negatively damped (Unstable) response if $\zeta < 0$



$$s_1 = -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$
$$s_2 = -\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1}$$

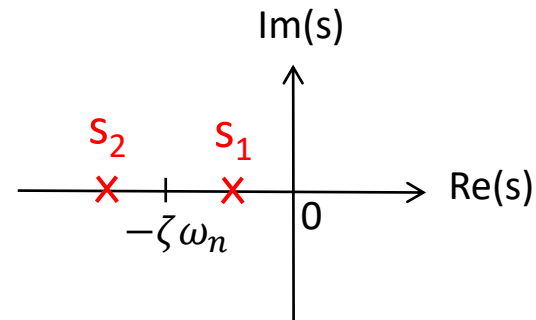
Second Order Systems

□ Damping Properties of Step Response

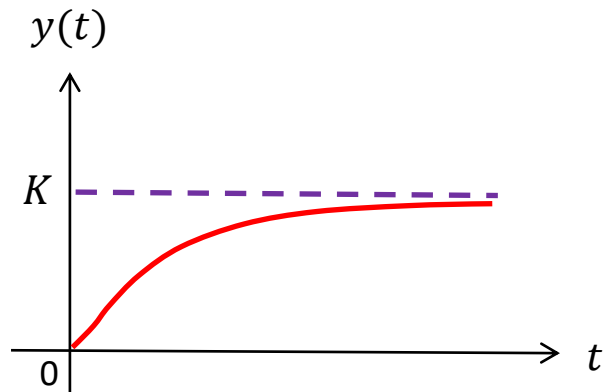
Overdamped System $\zeta > 1$

- System has two distinct real negative poles

$$s_{1,2} = -\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$$



- Step response is slow with no oscillation
- The response becomes slower by increasing ζ

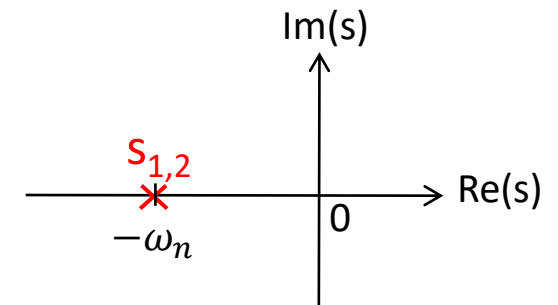


$$y(t) = K + C_1 e^{s_1 t} + C_2 e^{s_2 t}, \quad t \geq 0$$

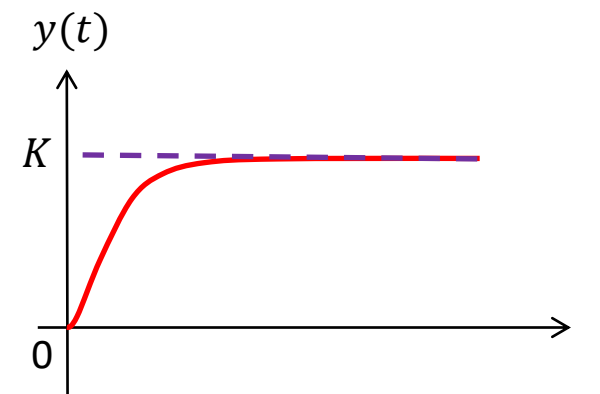
Critically-damped System $\zeta = 1$

- System has two repeated real negative poles

$$s_1 = s_2 = -\omega_n$$



- Step response has no oscillation
- Step response is always faster than the overdamped system



$$y(t) = K - K e^{-\omega_n t} (1 + \omega_n t), \quad t \geq 0$$

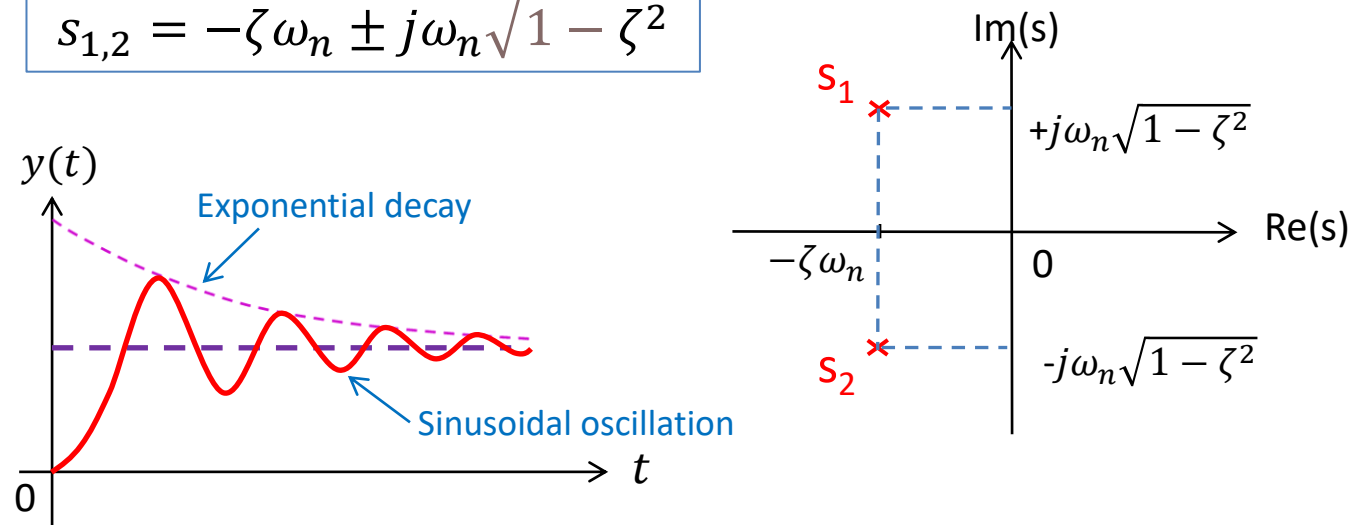
Second Order Systems

□ Damping Properties of Step Response

Underdamped System $0 < \zeta < 1$

- System has a pair of complex conjugated poles

$$s_{1,2} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$



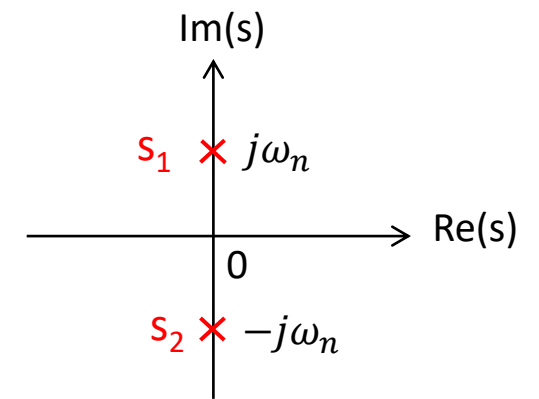
$$y(t) = K - \frac{Ke^{-\zeta\omega_n t}}{\sqrt{1-\zeta^2}} (\sin(\omega_d t + \cos^{-1}\zeta)), \quad t \geq 0$$

- Transient response has oscillations with frequency of $\omega_d = \omega_n\sqrt{1-\zeta^2}$
- ω_d is called damped natural frequency
- More oscillatory with larger overshoot by decreasing the ζ

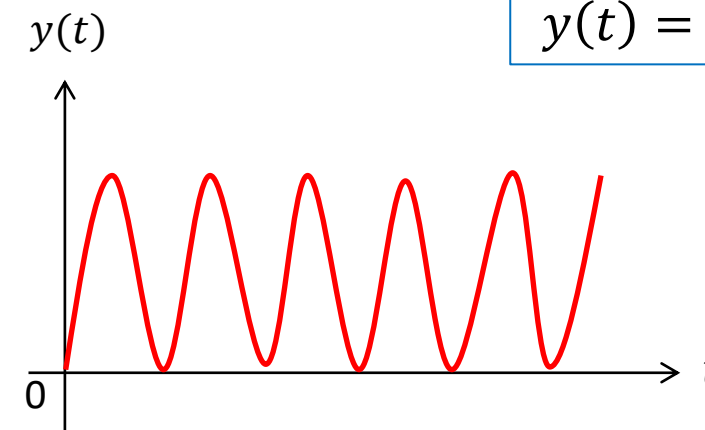
Undamped System $\zeta = 0$

- System has one pair of complex conjugate poles on the imaginary axes.

$$s_{1,2} = \pm j\omega_n$$



- Step response has sustained oscillation with frequency of ω_n



$$y(t) = K - K\cos(\omega_n t), \quad t \geq 0$$

Second Order Systems - Examples

□ Underdamped Response

In **underdamped** system, the damping force is not large enough to overcome the oscillation.

Assume $m = 50\text{kg}$, $b = 30\text{ Ns/m}$ and $k = 100\text{N/m}$ and the applied force is $f(t) = 100\text{N}$:

$$50\ddot{x}(t) + 30\dot{x}(t) + 100x(t) = f(t)$$

Transfer Function $\rightarrow G(s) = \frac{1}{50s^2 + 30s + 100} = \frac{0.02}{s^2 + 0.6s + 2}$

Characteristic Equation $\rightarrow s^2 + 0.6s + 2 = 0$

Undamped Natural Frequency $\rightarrow \omega_n^2 = 2 \rightarrow \omega_n = \sqrt{2} = 1.41\text{ rad/s}$

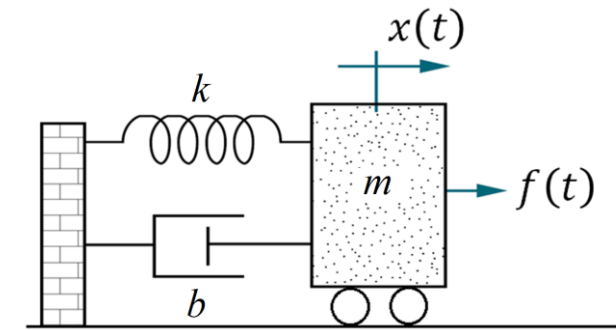
Damping Ratio $\rightarrow 2\zeta\omega_n = 0.6 \rightarrow \zeta = 0.21$

Poles: $s^2 + 0.6s + 2 = 0 \rightarrow s_{1,2} = -0.3 \pm j1.38$ (Complex poles)

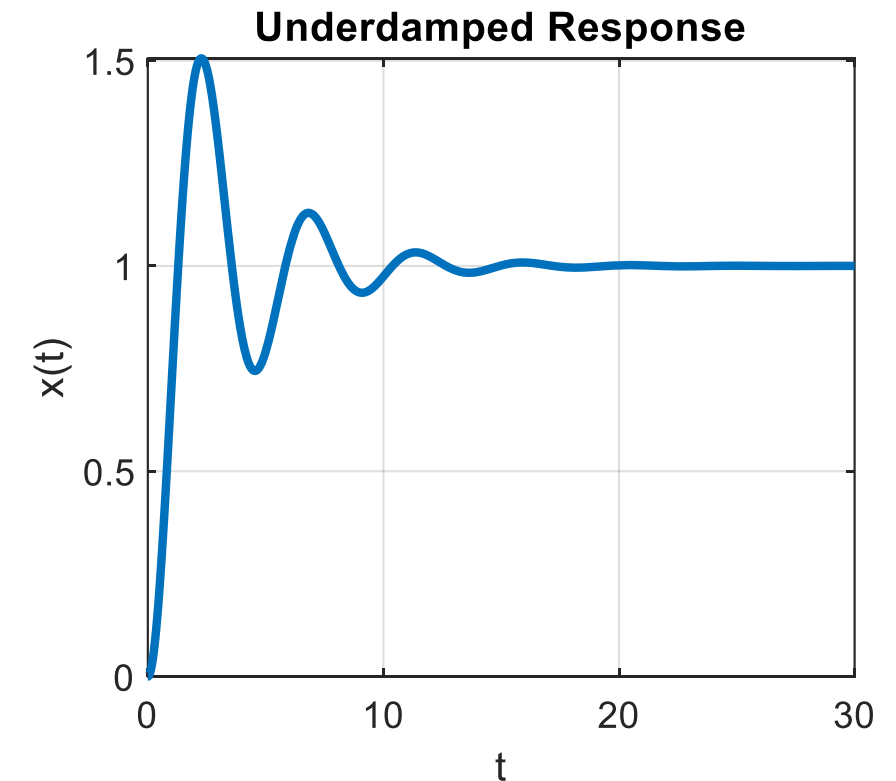
Step response: $x(t) = 1 - 1.02e^{-0.3t} \sin(1.38t + 77.88^\circ)$

Steady-state response
Transient Response

This solution shows that the mass oscillates with the **damped frequency** ω_d , and the transient response will disappear with time due to the **exponential** term.



$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$



Second Order Systems - Examples

Overdamped Response

In **over-damped** system, the damping force is large enough to limit the velocity and avoid the overshoot.

Assume $m = 50\text{kg}$, $b = 300\text{ Ns/m}$ and $k = 100\text{N/m}$ and the applied force is $f(t) = 100\text{N}$:

$$50\ddot{x}(t) + 300\dot{x}(t) + 100x(t) = f(t)$$

Transfer Function $\rightarrow G(s) = \frac{1}{50s^2 + 300s + 100} = \frac{0.02}{s^2 + 6s + 2}$

Characteristic Equation $\rightarrow s^2 + 6s + 2 = 0$

Undamped Natural Frequency $\rightarrow \omega_n^2 = 2 \rightarrow \omega_n = \sqrt{2} = 1.41\text{ rad/s}$

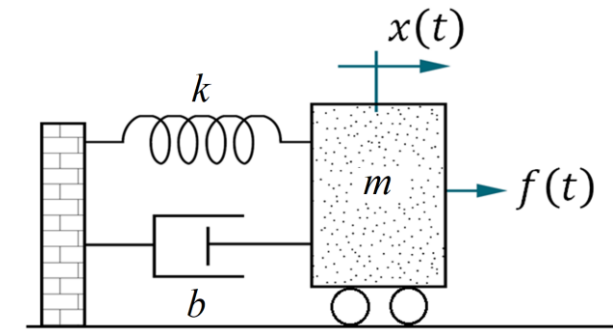
Damping Ratio $\rightarrow 2\zeta\omega_n = 6 \rightarrow \zeta = 2.12$

Poles: $s^2 + 6s + 2 = 0 \rightarrow s_1 = -5.65, s_2 = -0.35$ (Distinct, real poles)

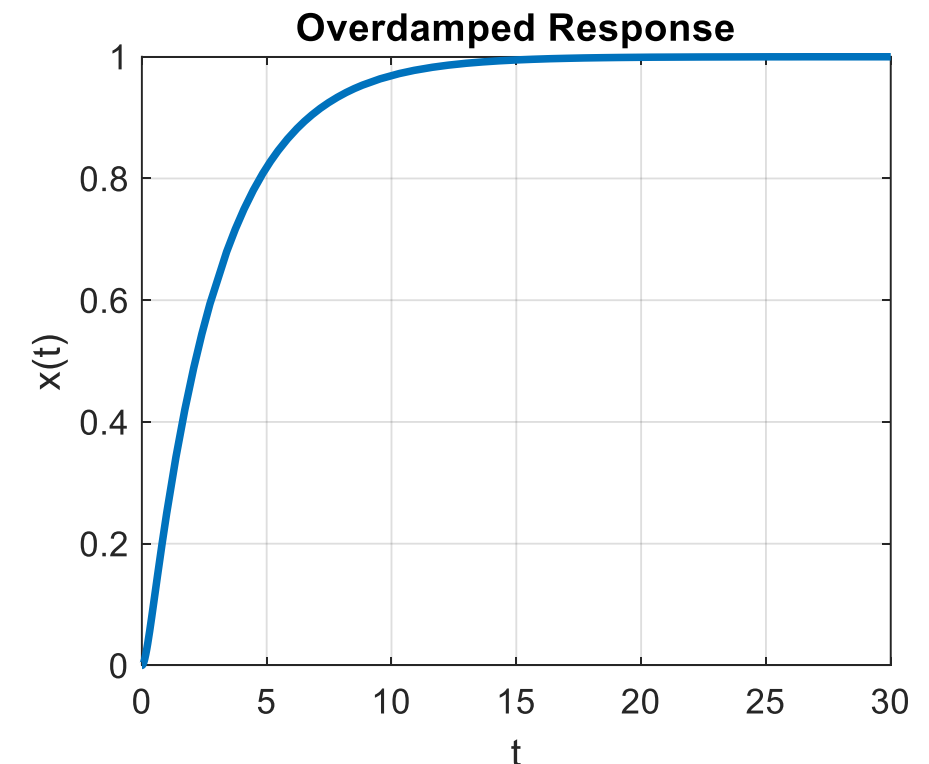
Step response: $x(t) = 1 + 0.067e^{-5.65t} - 1.067e^{-0.35t}$

Steady-state response
Transient Response

The transient response consists of exponential terms, and the steady-state response is 1.



$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$



Second Order Systems - Examples

□ Undamped Response (No damping, $b = 0$)

When there is no damping or friction force, the system will be a simple mass-spring system:

Assume $m = 50\text{kg}$, $b = 0$ and $k = 100\text{N/m}$ and the applied force is $f(t) = 100\text{N}$:

$$50\ddot{x}(t) + 100x(t) = f(t)$$

$$\text{Transfer Function} \rightarrow G(s) = \frac{1}{50s^2 + 100} = \frac{0.02}{s^2 + 2}$$

$$\text{Characteristic Equation} \rightarrow s^2 + 2 = 0$$

$$\text{Undamped Natural Frequency} \rightarrow \omega_n^2 = 2 \rightarrow \omega_n = \sqrt{2} = 1.41 \text{ rad/s}$$

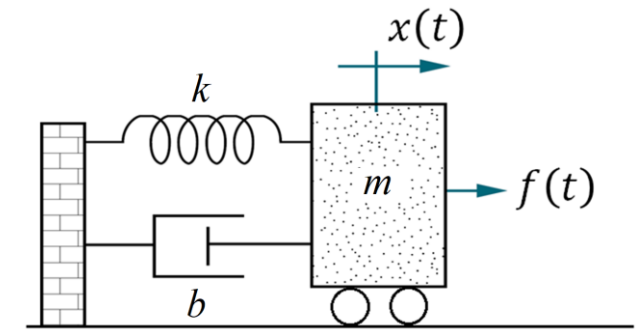
$$\text{Damping Ratio} \rightarrow 2\zeta\omega_n = 0 \rightarrow \zeta = 0$$

$$\text{Poles: } s^2 + 2 = 0 \rightarrow s = \pm j\sqrt{2} \quad (\text{Imaginary poles})$$

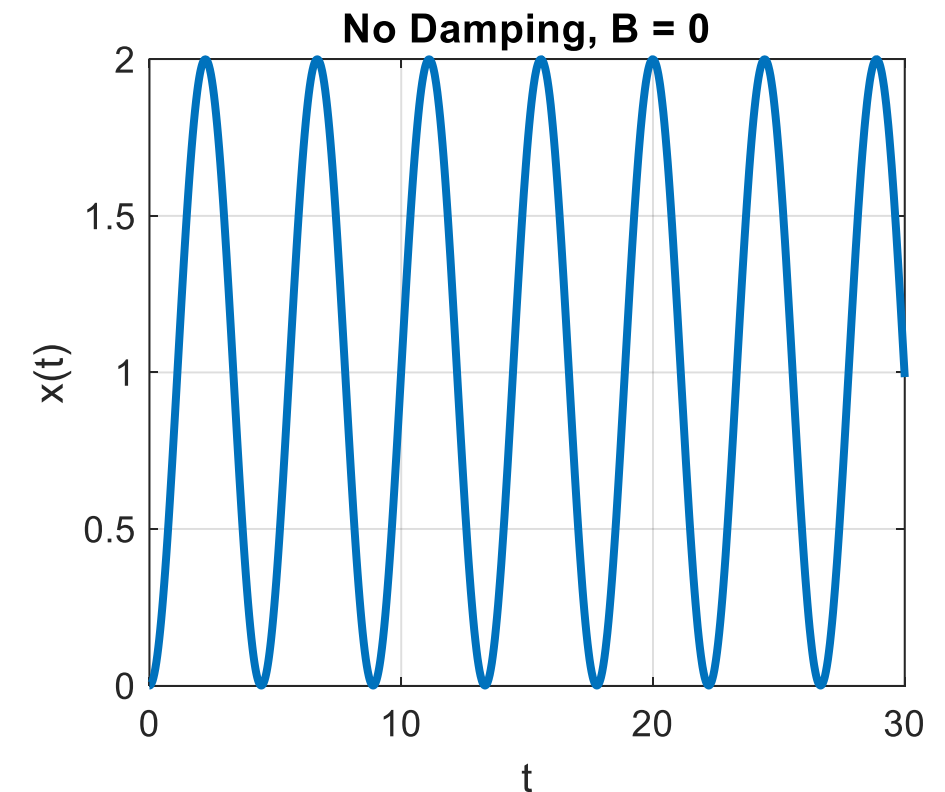
$$\text{Step response: } x(t) = 1 - \cos(\sqrt{2} t)$$

This solution shows that the mass oscillates with the **natural frequency** ω_n , and the transient response will not disappear with time.

$$\omega_n = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{50}} = \sqrt{2} \text{ rad/s}$$



$$m\ddot{x}(t) + b\dot{x}(t) + kx(t) = f(t)$$



Second-Order Model Identification from Step Response

□ Identification of Underdamped Systems

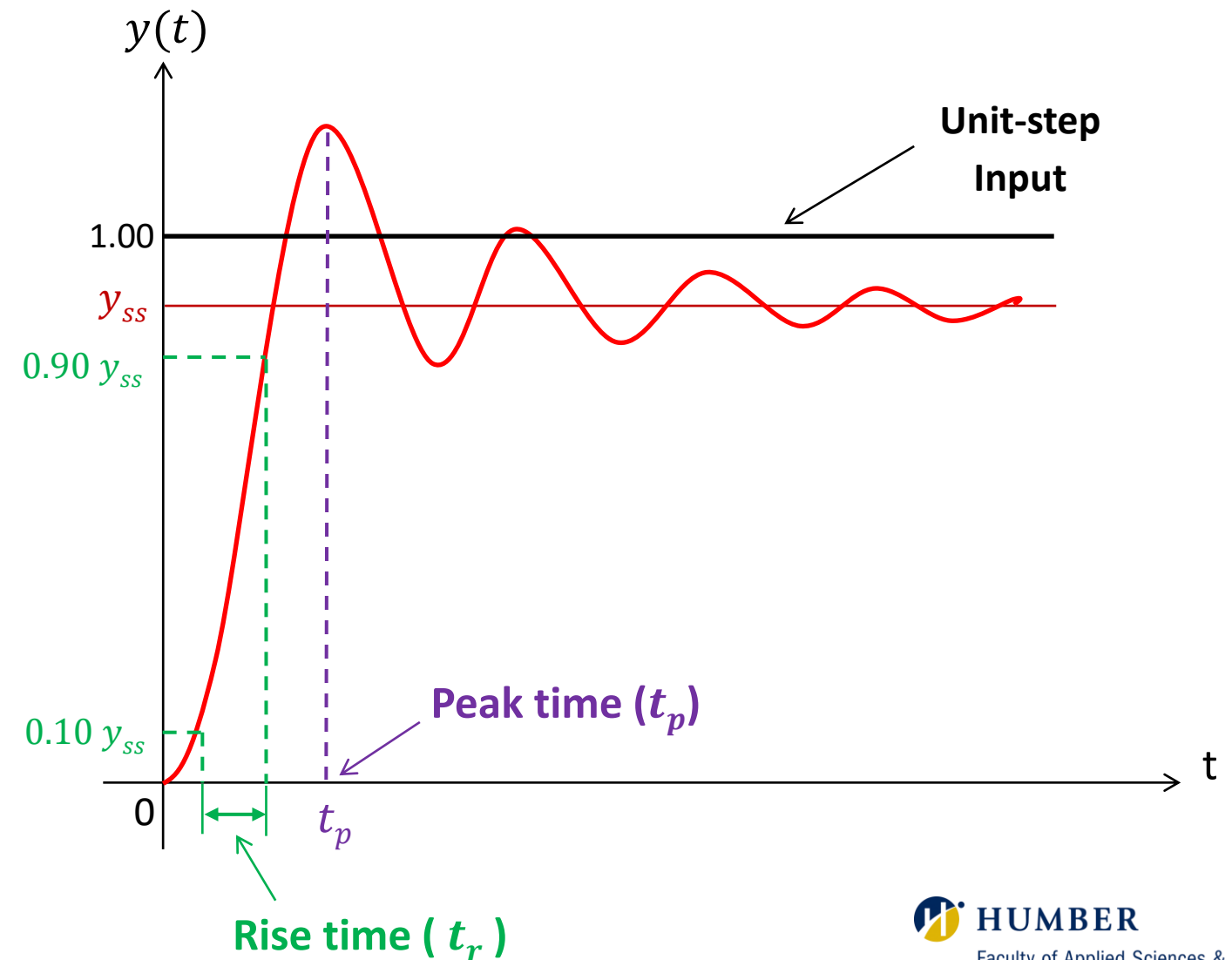
- Step response of an **underdamped system** ($0 < \zeta < 1$) is characterized by *oscillations*.
- Systems with **oscillatory step response** can be characterized by different **parameters** that can be determined from the step response.
- A number of such parameters are indicted below:

- **Rise time (t_r)**: The time required for the unit-step response to rise from 10% to 90% of its final value.

$$t_r \cong \frac{0.8 + 2.5\zeta}{\omega_n}$$

- **Peak time (t_p)**: The time required for the unit-step response to reach the first peak of the overshoot.

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}$$



Second-Order Model Identification from Step Response

□ Identification of Underdamped Systems

- Step response of an **underdamped system** ($0 < \zeta < 1$) is characterized by **oscillations**.
- Systems with **oscillatory step response** can be characterized by different **parameters** that can be determined from the step response.
- A number of such parameters are indicted below:

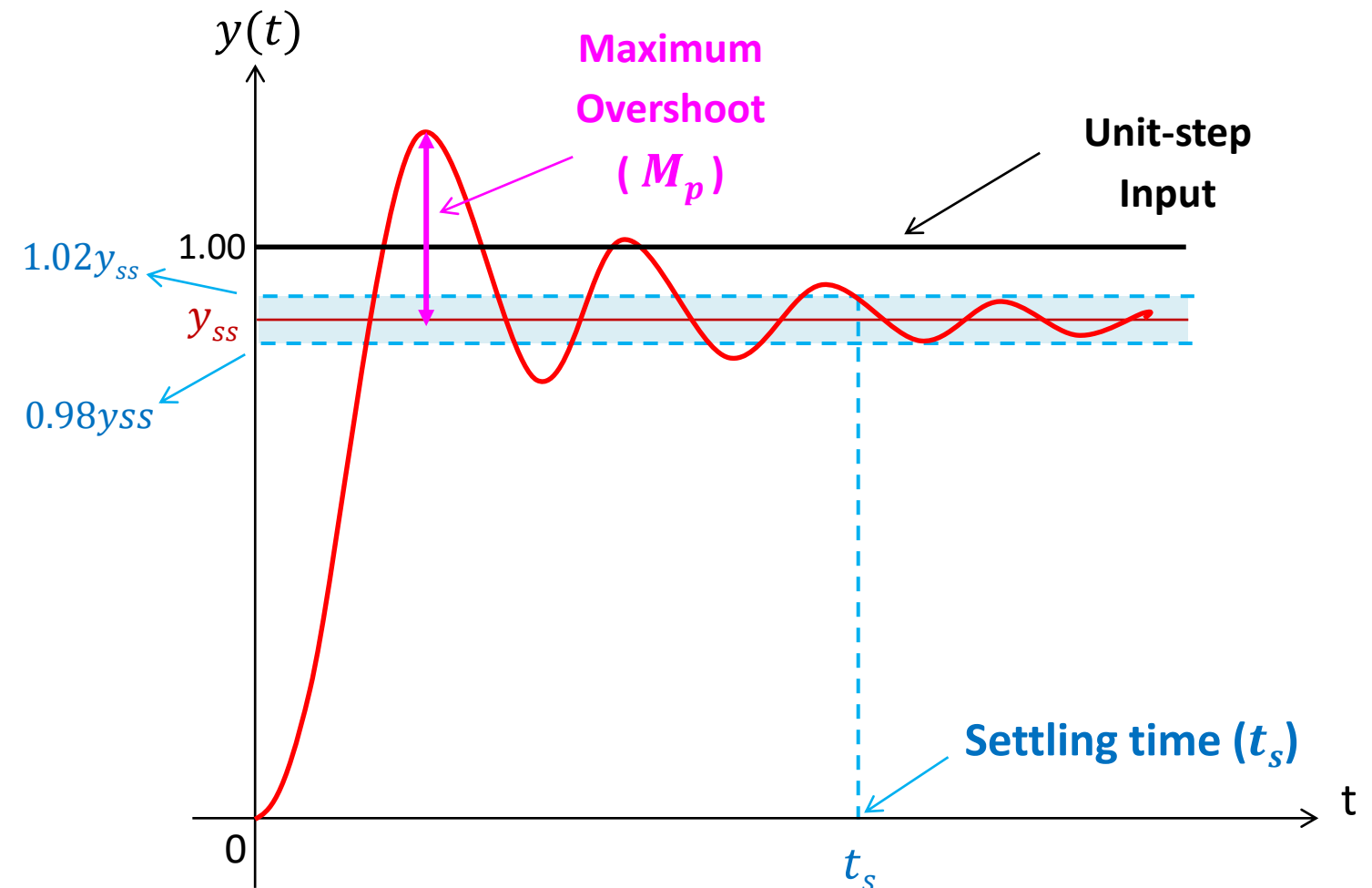
- **Maximum overshoot (M_p)**: The maximum peak value of the unit-step response measured from the final value of the response.

$$\%O.S. = \frac{M_p}{y_{ss}} = \frac{y(t_p) - y_{ss}}{y_{ss}} \times 100\%$$

$$O.S. = e^{-\zeta\pi/\sqrt{1-\zeta^2}}$$

- **Settling time (t_s)**: The time required for the unit-step response to reach and stay within the specified percentage of its final value (usually 2%)

$$2\% \text{ criteria} \rightarrow t_s \approx \frac{4}{\zeta\omega_n}, \quad 0 < \zeta < 0.9$$



Second-Order Model Identification from Step Response

□ Identification of Underdamped Systems

- The followings are the steps to complete identification of an **underdamped** system

- 1) Measure y_{ss} and $y(t_p)$ and compute the **damping ratio** ζ

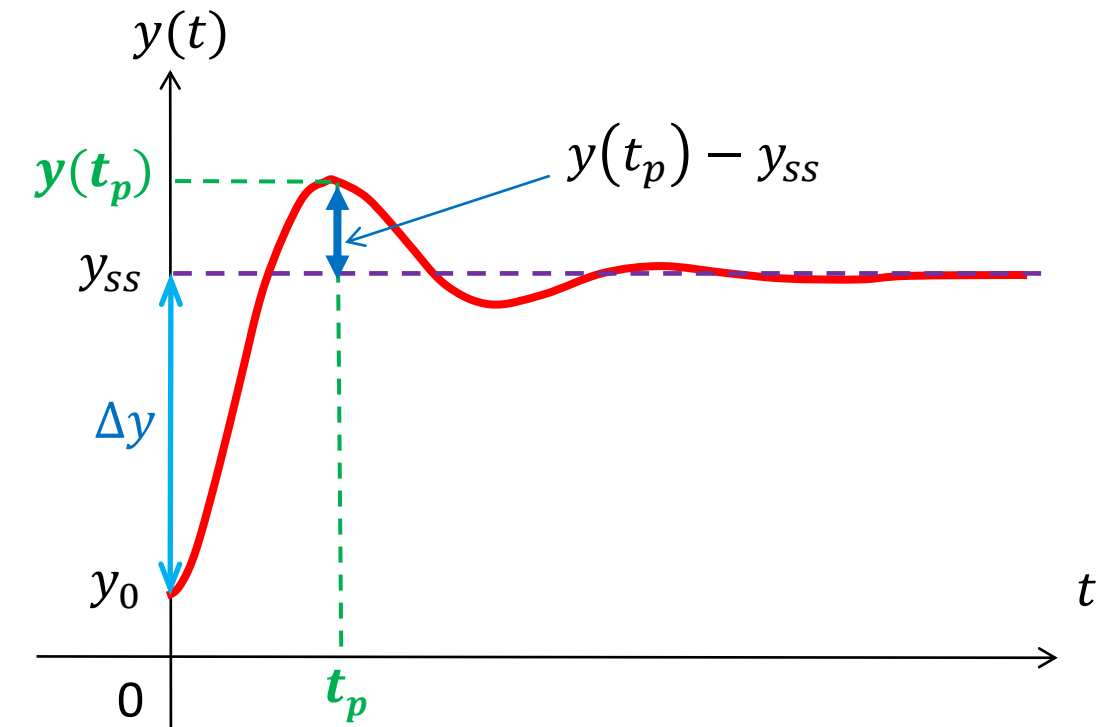
$$O.S. = \frac{y(t_p) - y_{ss}}{y_{ss} - y_0} \rightarrow \boxed{\zeta = \frac{-\ln(O.S.)}{\sqrt{\pi^2 + \ln^2(O.S.)}}}$$

- 2) Measure the peak time t_p and compute the **natural frequency** ω_n

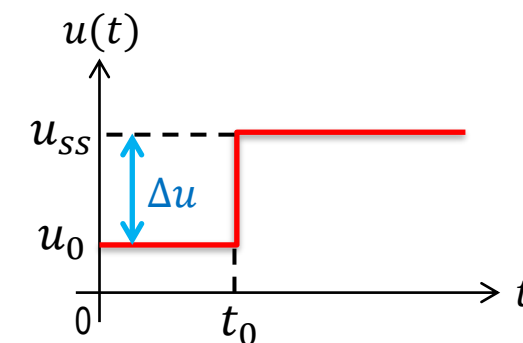
$$\boxed{t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}}}$$

- 3) The **static gain** K is obtained with the previously explained approach in the first-order system models.

$$\boxed{K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0}}$$



$$G(s) = \frac{K \omega_n^2}{s^2 + 2\zeta \omega_n s + \omega_n^2}$$



Second-Order Model Identification from Step Response

Example 3

Determine a second-order transfer function model of the system based on the given unit-step response.

First, find the **damping ratio**:

$$O.S. = \frac{y(t_p) - y_{ss}}{y_{ss} - y_0} = \frac{0.9 - 0.7}{0.7 - 0} = 0.29$$

$$\zeta = \frac{-\ln(O.S.)}{\sqrt{\pi^2 + \ln^2(O.S.)}} = \frac{-\ln(0.29)}{\sqrt{\pi^2 + \ln^2(0.29)}} \rightarrow \boxed{\zeta = 0.367}$$

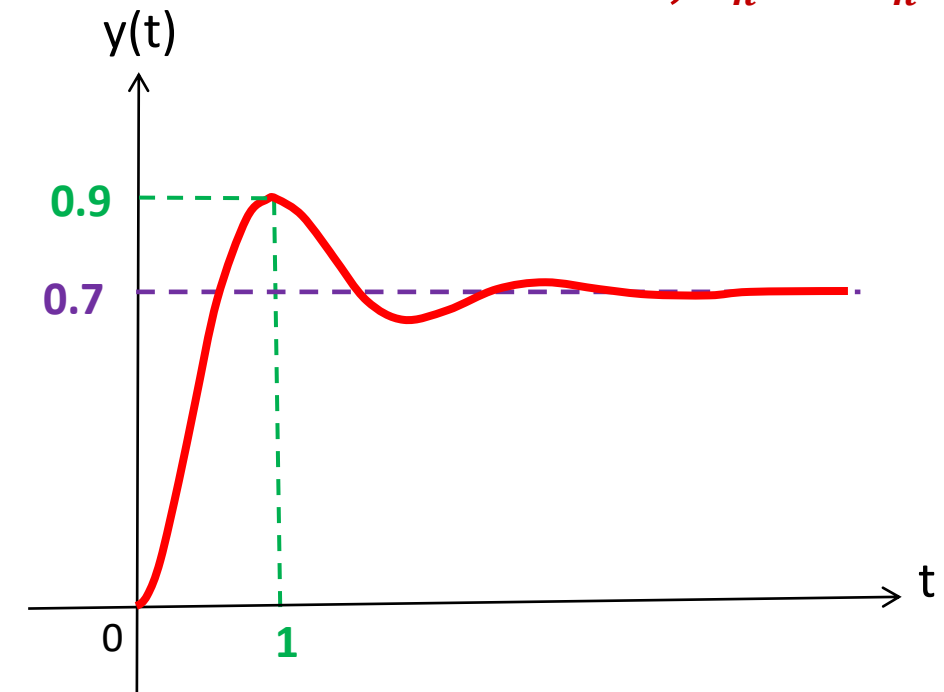
The **undamped natural frequency** is calculated as below

$$t_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} \rightarrow 1 = \frac{\pi}{\omega_n \sqrt{1 - (0.367)^2}} \rightarrow \boxed{\omega_n = 3.38 \text{ rad/s}}$$

The **static gain** is computed as below

$$K = \frac{\Delta y}{\Delta u} = \frac{y_{ss} - y_0}{u_{ss} - u_0} = \frac{0.7 - 0}{1 - 0} \rightarrow \boxed{K = 0.7}$$

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$



The **second-order model** is obtained as:

$$G(s) = 0.7 \frac{11.43}{s^2 + 2.48s + 11.43}$$

Second-Order Model Identification from Step Response

Example 3

Determine a second-order transfer function model of the system based on the given unit-step response.

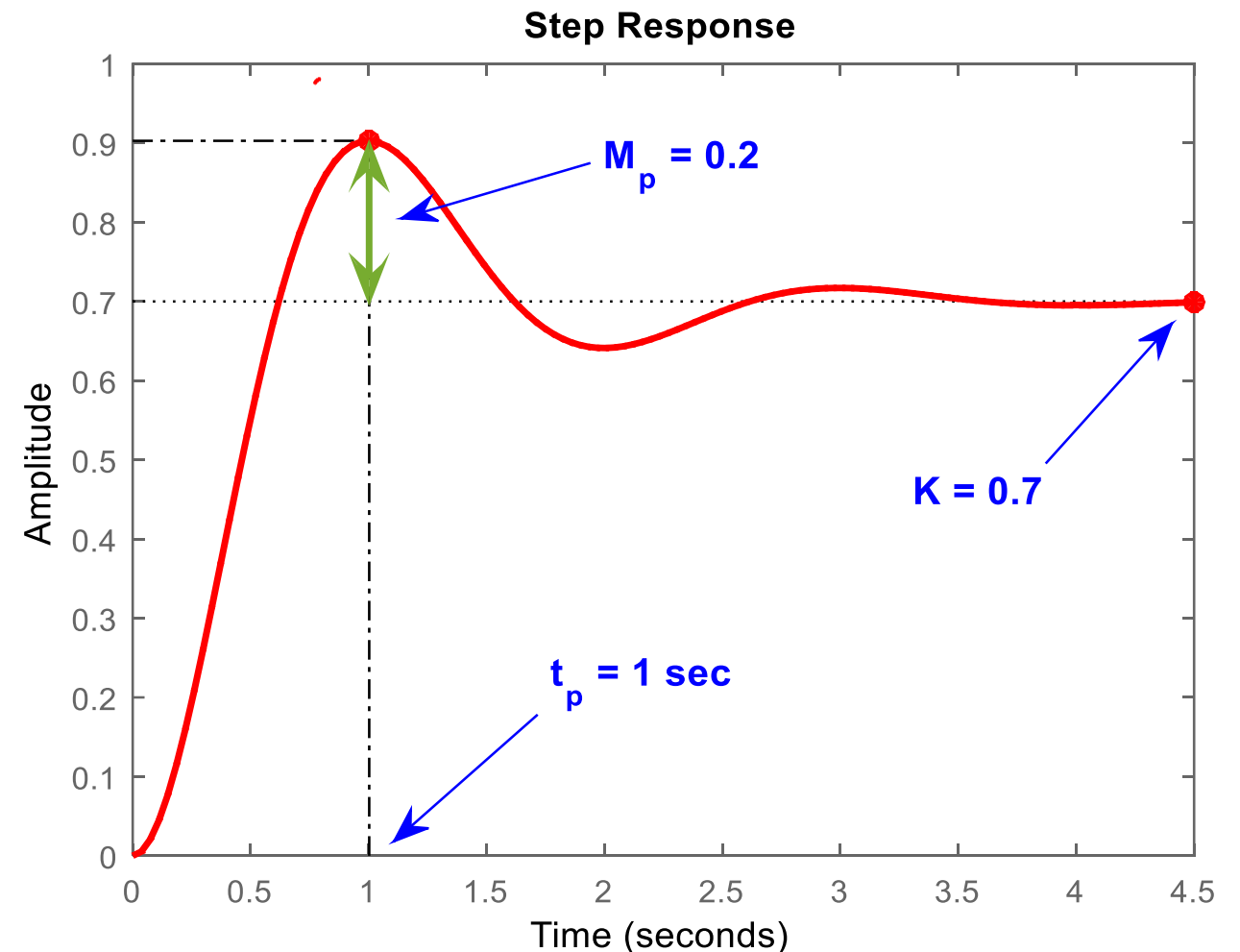
Model Verification: We can plot unit-step responses of the second-order model by MATLAB to compare the time response specifications them.

$$K = 0.7, \quad t_p = 1 \text{ sec}, \quad M_p = 0.9 - 0.7 = 0.2$$

$$G(s) = 0.7 \frac{11.43}{s^2 + 2.48s + 11.43}$$

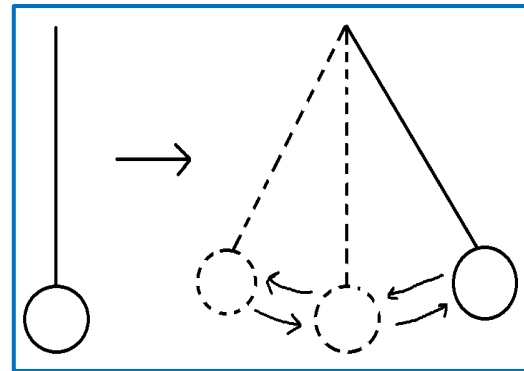
```
num = 0.7*[11.43];  
den = [1 2.48 11.43];  
sys = tf(num,den);  
  
stepplot(sys)
```

$$G(s) = \frac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

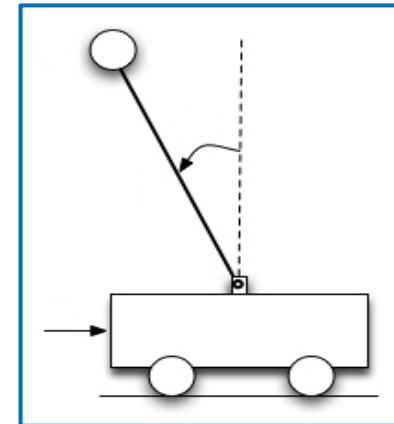


Stability of Systems

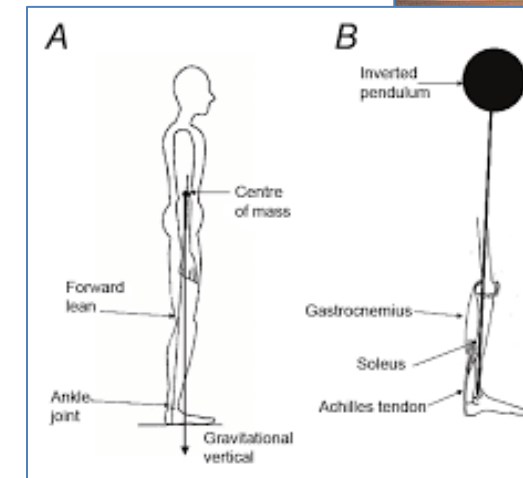
- In general, a **stable** system is the one that will
 - **Remain at rest** unless excited by an external source
 - **Return to rest** if all excitations are removed
- **Simple pendulum** hanging from the ceiling is a **stable** system, but an **inverted pendulum** is an **unstable** system.



Simple Pendulum



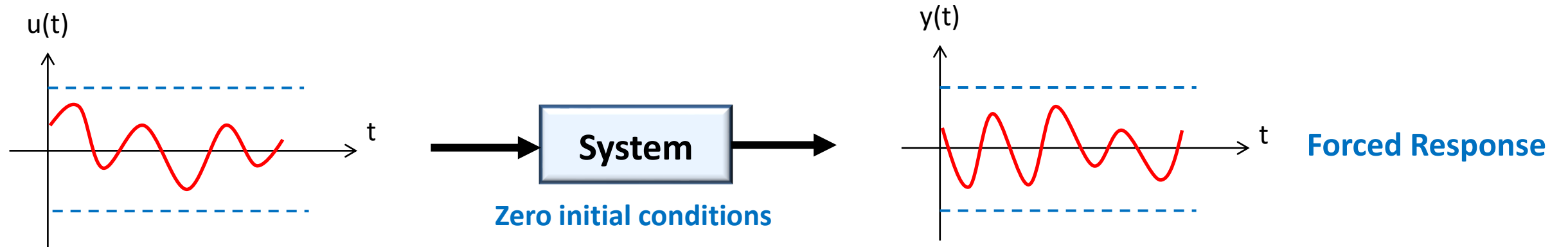
Inverted Pendulum



Stability of Systems

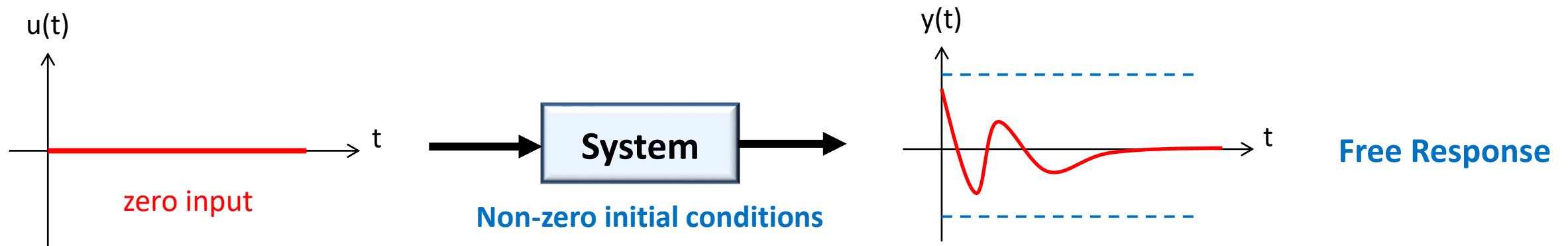
□ Bounded-Input, Bounded-Output Stability (BIBO Stability)

- A system with **zero initial conditions** is **BIBO stable** if every bounded-input results in a bounded-output.



□ Asymptotic Stability (Zero-Input Stability)

- A system with **zero input** is **asymptotically stable** if following conditions are satisfied:



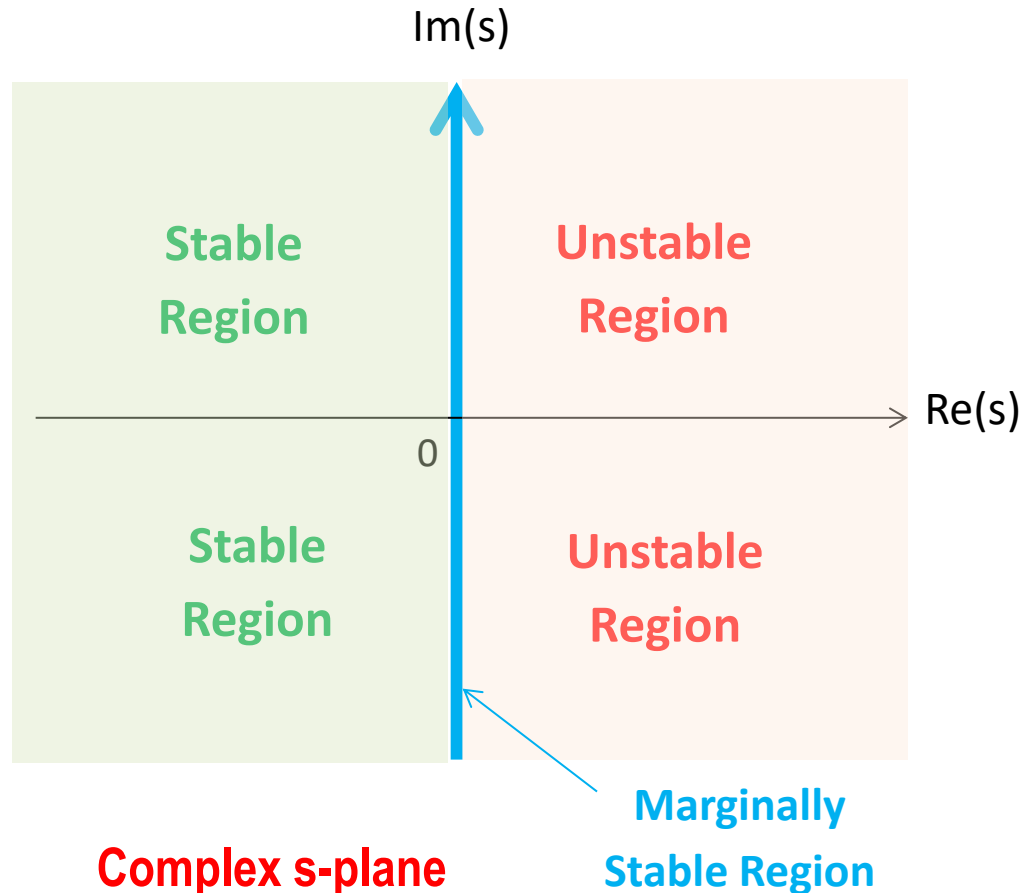
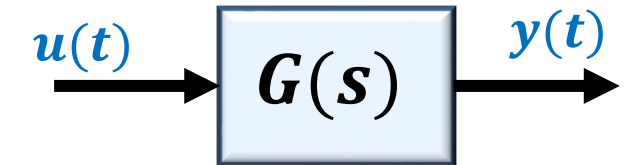
- In **LTI systems** the **asymptotic stability** is equivalent to **BIBO stability**, but in general they are different.

Stability of Systems

□ Stability of Linear Dynamic Systems

- Consider the a LTI dynamic system with input $u(t)$, output $y(t)$ and transfer function model of $G(s)$

A linear dynamic system is **stable** if and only if all **poles** of $G(s)$ are in the **left-half s-plane**



Example 4

$$G_1(s) = \frac{s + 1}{(s + 2)(s^2 + 5)}$$

$$\text{poles} \rightarrow \begin{cases} s_1 = -2 \\ s_{2,3} = \pm j\sqrt{5} \end{cases} \quad \text{Marginally Stable}$$

$$G_2(s) = \frac{2(s + 2)}{(s + 10)(s + 3)}$$

$$\text{poles} \rightarrow \begin{cases} s_1 = -10 \\ s_2 = -3 \end{cases} \quad \text{Stable}$$

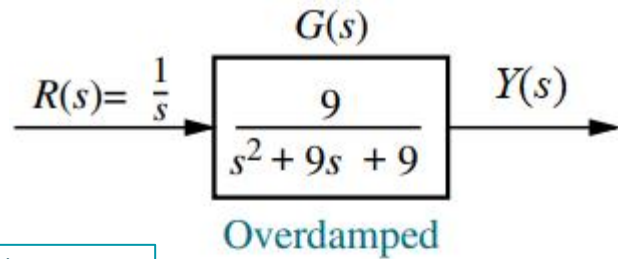
$$G_3(s) = \frac{10}{(s - 10)(s^2 + 4)}$$

$$\text{poles} \rightarrow \begin{cases} s_1 = 10 \\ s_2 = \pm j2 \end{cases} \quad \text{Unstable}$$

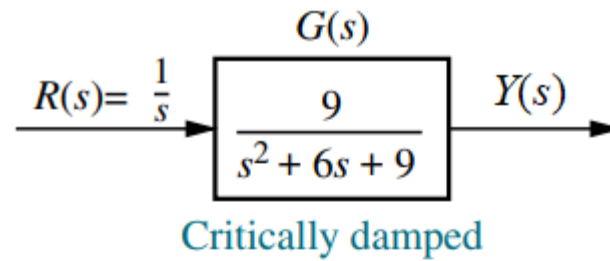
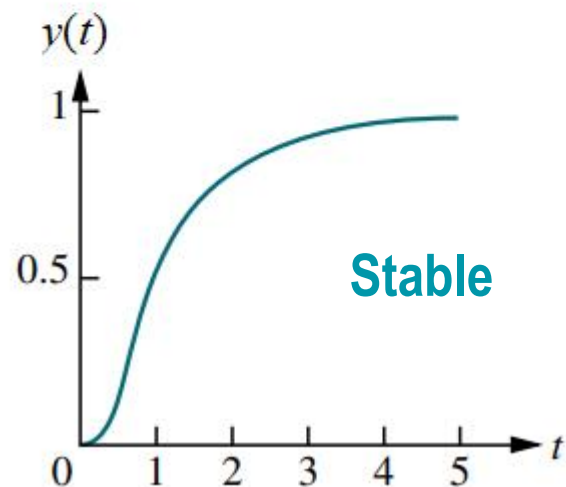
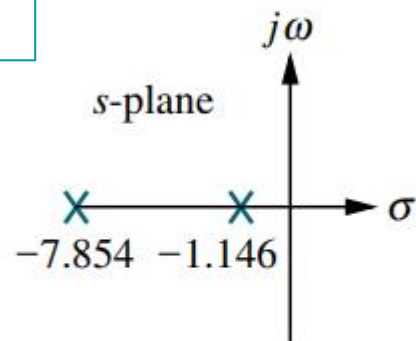
Step Response, Pole Locations & Stability

Example 5

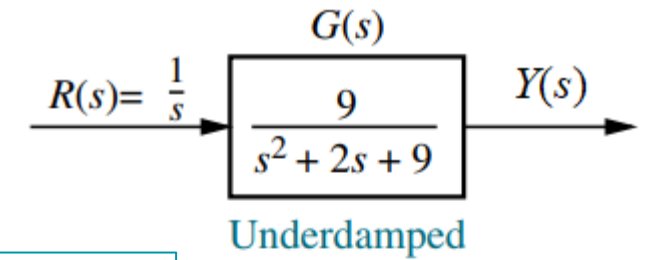
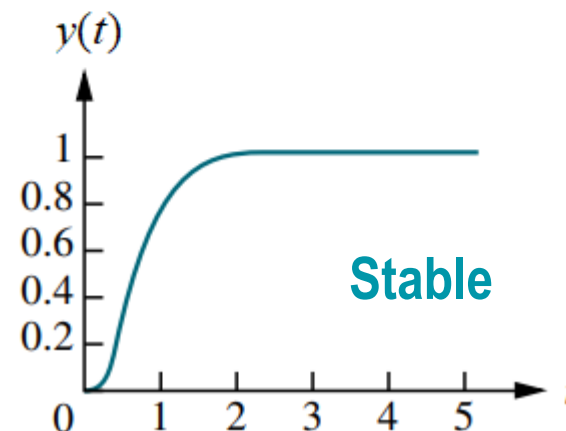
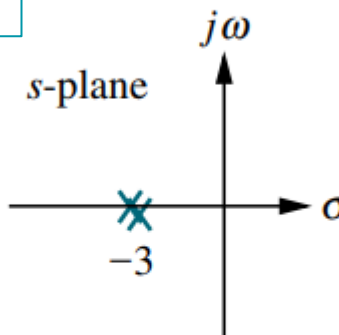
This example shows the relationship of the step response, pole locations and stability in a second-order system.



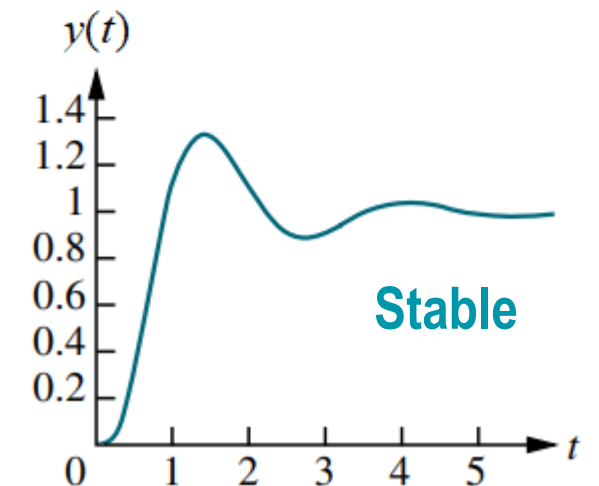
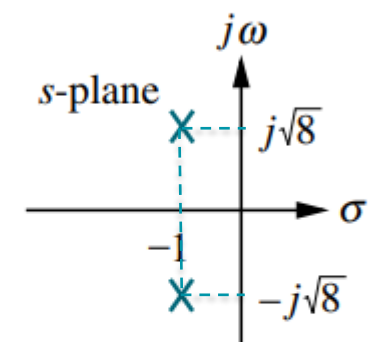
$$\zeta = 1.5$$
$$s_1 = -7.854,$$
$$s_2 = -1.146$$



$$\zeta = 1$$
$$s_1 = s_2 = -3$$



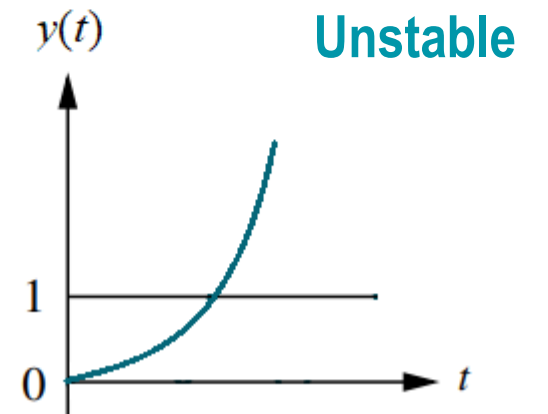
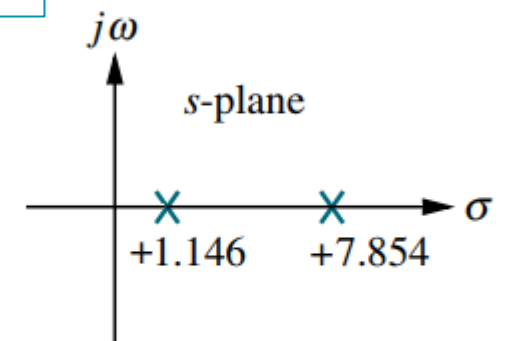
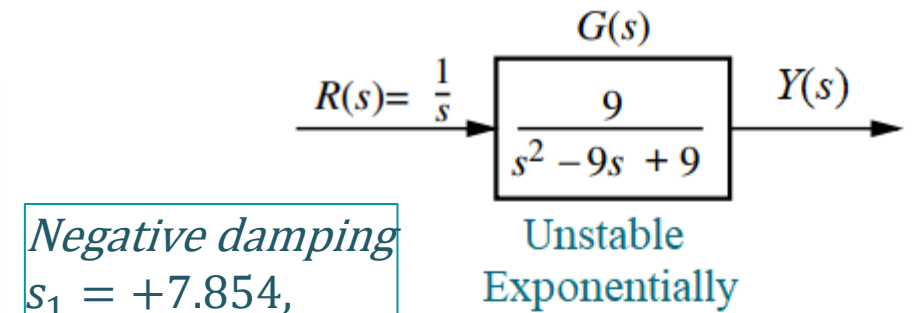
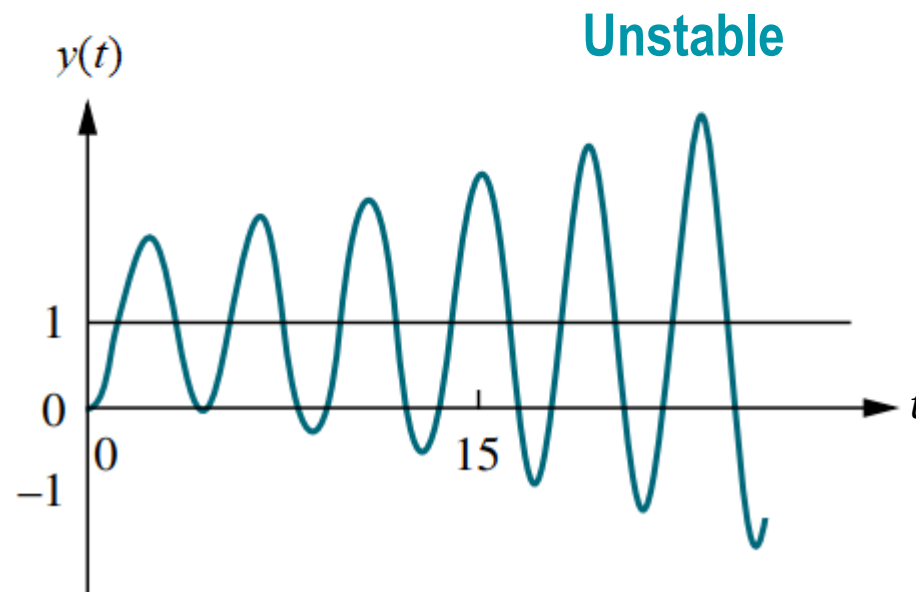
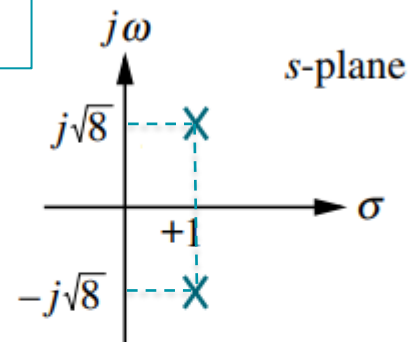
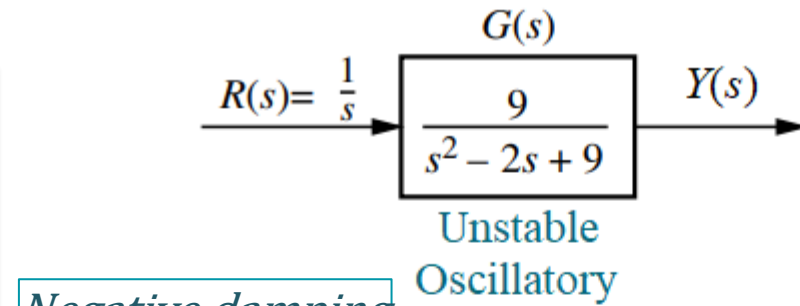
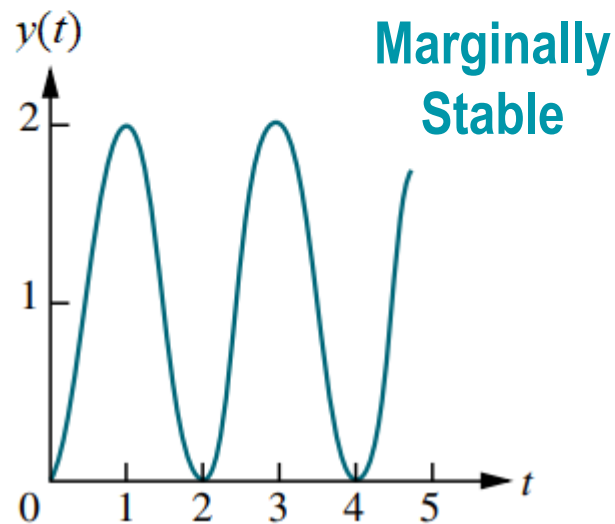
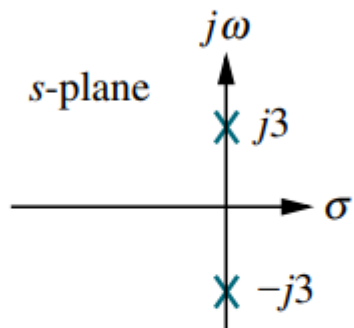
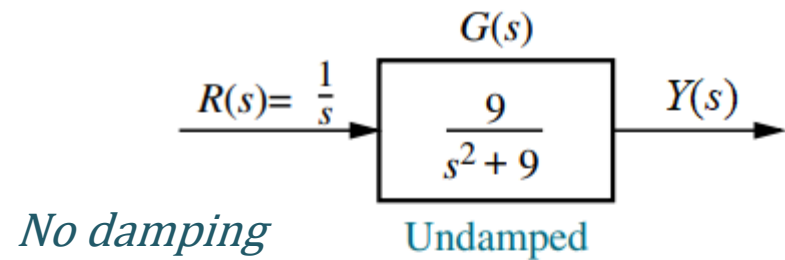
$$\zeta = 0.33$$
$$s_{1,2} = -1 \pm j\sqrt{8}$$



Step Response, Pole Locations & Stability

Example 5

This example shows the relationship of the step response, pole locations and stability in a second-order system.



Low-Order Model Approximation

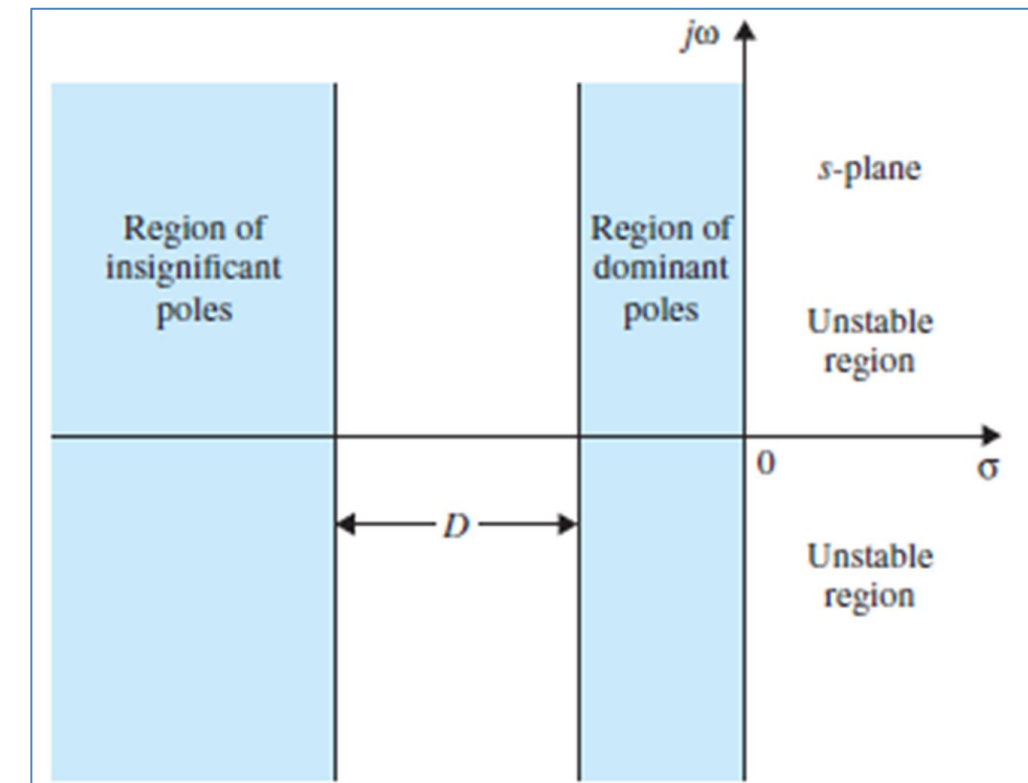
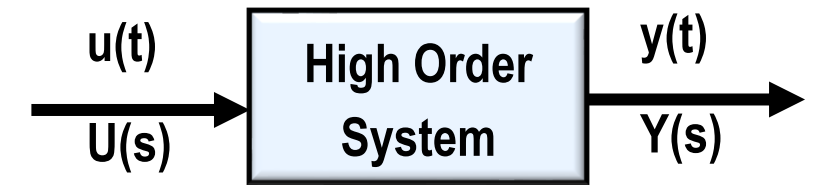
□ Low-Order Models & Dominant Poles

- The transfer function of a high-order LTI system can be approximated by a **lower-order model** one by eliminating some **insignificant poles** and **zeroes**.

$$G(s) = \frac{Y(s)}{U(s)} = \frac{K(s + z_1)(s + z_2) \cdots (s + z_m)}{(s + p_1)(s + p_2) \cdots (s + p_r) \cdots (s^2 + 2\zeta\omega_n s + \omega_n^2)}$$

$$G(s) = \frac{A_1}{s + p_1} + \cdots + \frac{A_r}{s + p_r} + \cdots + \frac{Bs + C}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

- The poles **close to the imaginary axis** are considered as the **Dominant Poles** of the system. Because, they have **larger time constant** and have more effect on the transient response.
- The **stable poles** are located **very far from the origin** (10 times farther from **dominant poles**) have low effect on the transient response and may be neglected.
- A pair of closely located **stable poles** and **stable zeros** can effectively cancel each other, if the **residue** of the pole is **much smaller** than the other poles.



Low-Order Model Approximation

Example 6

Determine a low-order approximation for the following high-order system.

$$G(s) = \frac{50s + 275}{s^3 + 106s^2 + 605s + 500}$$

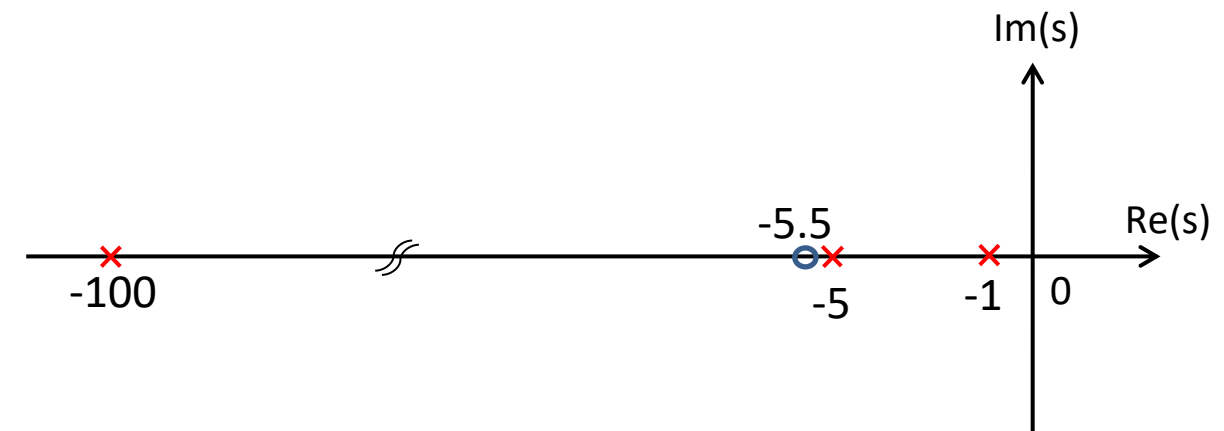
First, we have to find the **steady-state gain** and **pole-zero locations** for this system.

$$G(s) = \frac{50s + 275}{s^3 + 106s^2 + 605s + 500} = \frac{50(s + 5.5)}{(s + 1)(s + 5)(s + 100)}$$

Steady-state gain $\rightarrow G(0) = \frac{50 \times 5.5}{1 \times 5 \times 100} = 0.55$

Poles $\rightarrow p_1 = -1, p_2 = -5, p_3 = -100$

Zeros $\rightarrow z_1 = -5.5$



- The pole-zero pair of $p_2 = -5, z_1 = -5.5$ are both **stable** and **close to each other**, can be **canceled** if the residue is small enough.
- The pole $p_3 = -100$ is **very far from the origin**, so it can be **neglected**.
- The pole $p_1 = -1$ is **the dominant pole**.

$$G(s) = \frac{50(s + 5.5)}{(s + 1)(s + 5)(s + 100)} = \frac{0.568}{s + 1} + \frac{-0.066}{s + 5} + \frac{-0.502}{s + 100} \quad \Rightarrow \quad \boxed{G(s) \cong \frac{0.568}{s + 1}}$$

Since the residue of the pole at -5 is much smaller than the other poles, the **pole-zero cancellation is valid**.

Low-Order Model Approximation

Example 6

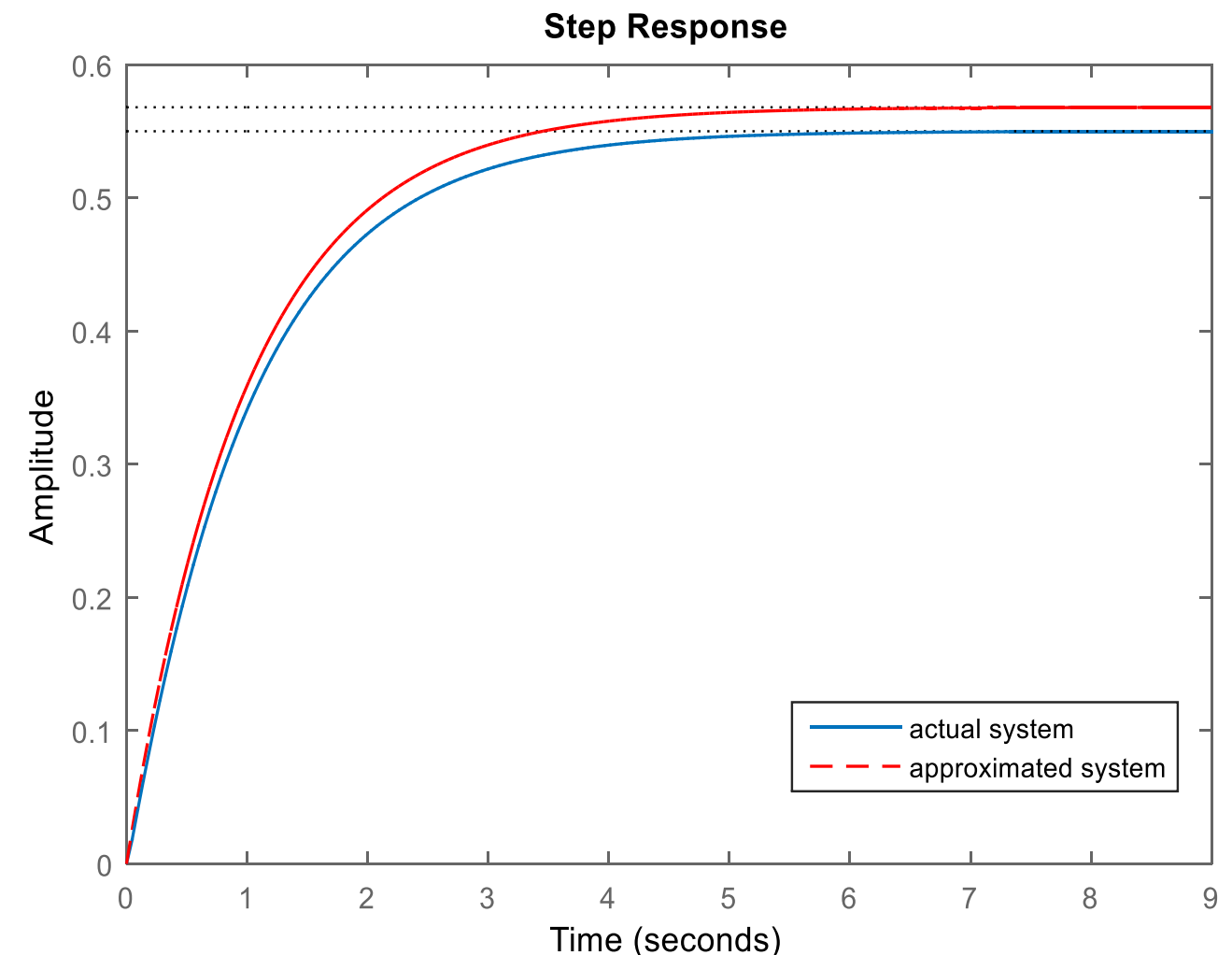
Determine a low-order approximation for the following high-order system.

$$G(s) = \frac{50s + 275}{s^3 + 106s^2 + 605s + 500}$$

Model Verification: We can plot unit-step responses of the original system and its approximated version by MATLAB to compare them.

$$G(s) \cong \frac{0.568}{s + 1}$$

```
num1 = [50 275];  
den1 = [1 106 605 500];  
sys1 = tf(num1,den1);  
  
num2 = [0.568];  
den2 = [1 1];  
sys2 = tf(num2,den2);  
  
step(sys1)  
hold on  
step(sys2)
```



THANK YOU